

尚硅谷大数据技术之 Git

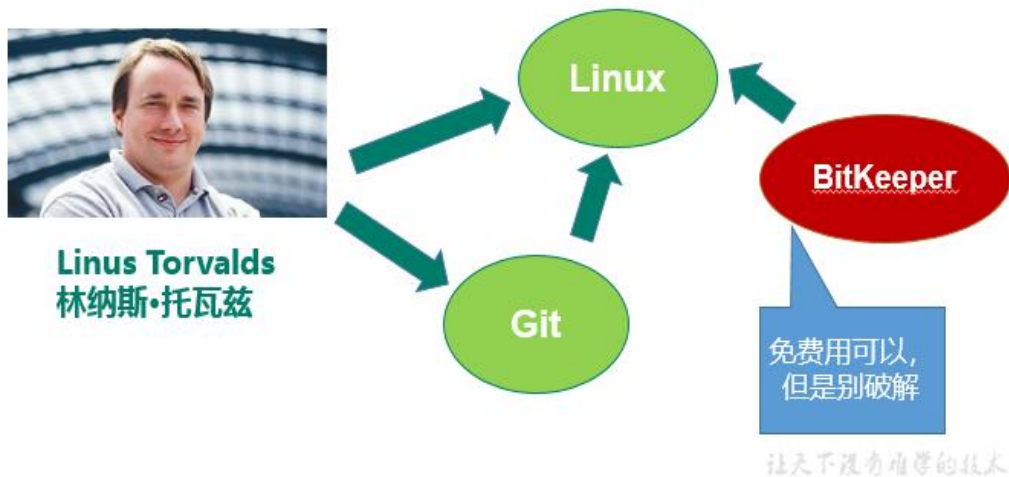
第 1 章 Git 简介

1.1 为什么要学习 Git

必点天赋 必然趋势

1.2 Git 是什么

Git 是目前世界上最先进的分布式版本控制系统。



注:BitKeepper 是非自由软件, 初期 Linux 核心开发是基于 BitKeeper 进行管理。

1.3 Git 能干什么

Git能干什么

尚硅谷

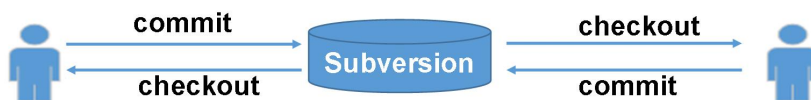


让天下没有难学的技术

1.4 集中管理型版本控制工具



尚硅谷



让天下没有难学的技术

1) 优点

- (1) 实现了大部分开发中对版本管理的需求
- (2) 结构简单，上手容易。

2) 缺点

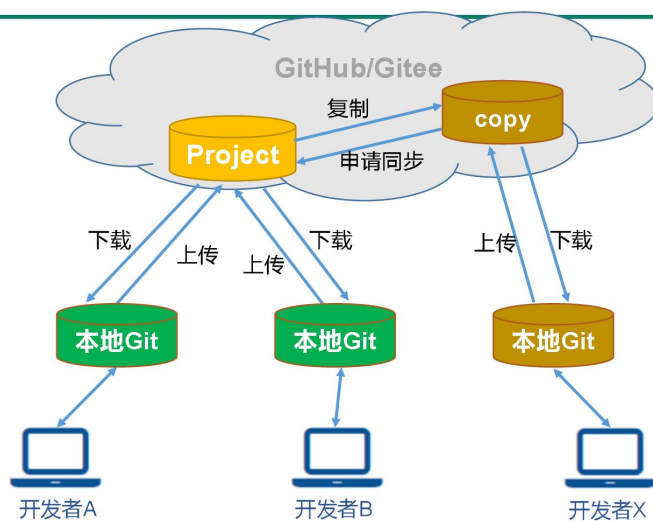
- (1) 版本管理的服务器一旦崩溃，硬盘损坏，代码无法恢复。
- (2) 程序员上传到服务器的代码要求是完整版本，无法完成程序员开发过程中想做小

版本的管理。

(3) 系统正在上线运行，时不时还要修改 bug，要增加好几个功能要几个月，无法同时管理多个版本。

(4) 无法管理一个分布在世界各地、互不相识的大型开发团队

1.5 分布式版本控制工具



让天下没有难学的技术

1) 优点:

- (1) 容灾能力强
- (2) 本地版本管理
- (3) 异地协作
- (4) 灵活分支

2) 缺点:

- (1) GitHub 是国外的网站,访问比较困难.可以使用国内的 gitee 来代替.

第 2 章 Git 的下载与安装

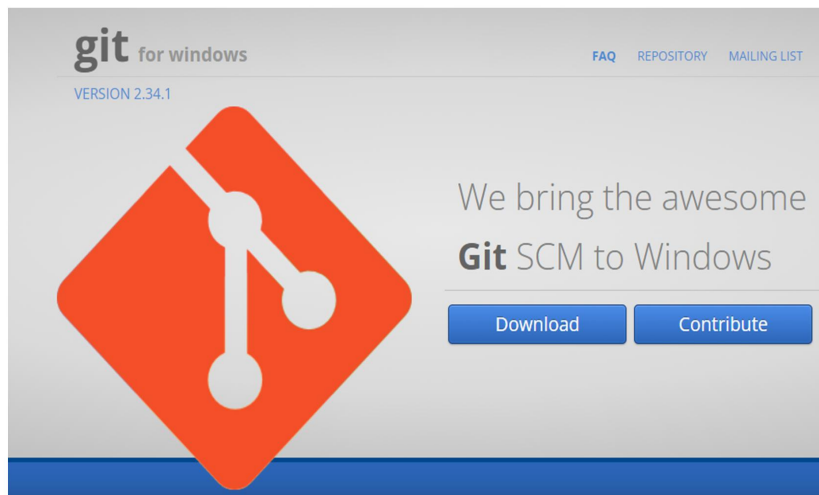
2.1 Git for windows

1) 下载地址:

<https://git-for-windows.github.io/>

<https://git-scm.com/download>

2) 安装



第 3 章 Git 实战操练

3.1 初始化版本库

1) 在项目文件夹内，执行 `git init`

例如：在 D 盘中创建 git 文件夹，在 git 文件夹中创建 gittest 项目，进入到 gittest 中，右键打开 git bash,执行 `git init`,执行完后会多出一个.git 目录。

```
Administrator@WIN-F38TFIC4EUV MINGW64 /d/git/gittest
$ git init
Initialized empty Git repository in D:/git/gittest/.git/
```

3.2 配置签名

1) 工作中建议配置企业邮箱，学习中建议配置的邮箱为 `gitHub` 或者 `gitee` 的注册邮箱

2) 配置当前项目作用域的签名 `user.name` 和 `user.email`

```
Administrator@WIN-F38TFIC4EUV MINGW64 /d/git/gittest (master)
$ git config user.name "zhangsan"

Administrator@WIN-F38TFIC4EUV MINGW64 /d/git/gittest (master)
$ git config user.email "zhangsan@atguigu.com"
```

3) 配置全局作用域签名


```
Administrator@WIN-F38TFIC4EUV MINGW64 /d/git/gittest (master)
$ git config --global user.name "weiyunhui"
g
Administrator@WIN-F38TFIC4EUV MINGW64 /d/git/gittest (master)
$ git config --global user.email "weiyunhui@atguigu.com"
```

4) 签名配置的存储位置

当前项目作用域签名配置的存储文件: 当前项目\ .git 目录下的 config 文件.

全局作用域签名配置的存储文件: C:\Users\当前系统用户名\目录下的 .gitconfig 文件

3.3 提交文件

1) 新建文件后, 通过 git status 进行查看文件状态

文件名显示红色表示还未纳入版本管理

```
Administrator@WIN-F38TFIC4EUV MINGW64 /d/git/gittest (master)
$ mkdir src
Administrator@WIN-F38TFIC4EUV MINGW64 /d/git/gittest (master)
$ cd src
Administrator@WIN-F38TFIC4EUV MINGW64 /d/git/gittest/src (master)
$ vim Test.java
Administrator@WIN-F38TFIC4EUV MINGW64 /d/git/gittest/src (master)
$ cd ..
Administrator@WIN-F38TFIC4EUV MINGW64 /d/git/gittest (master)
$ git status
On branch master

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
  src/

nothing added to commit but untracked files present (use "git add" to track)
```

2) 将文件添加到暂存区: git add 文件名 或者 git add 文件

```
Administrator@WIN-F38TFIC4EUV MINGW64 /d/git/gittest (master)
$ git add src/Test.java
warning: LF will be replaced by CRLF in src/Test.java.
The file will have its original line endings in your working directory
```

3) 再次 git status 查看文件状态

文件名显示绿色表示文件进入暂存区

```
Administrator@WIN-F38TFIC4EUV MINGW64 /d/git/gittest (master)
$ git status
On branch master

No commits yet

changes to be committed:
  (use "git rm --cached <file>..." to unstage)
  new file:   src/Test.java
```

4) 提交暂存区文件到本地库

git commit ,自动进入到 vim 中, 编写注释 ,完成提交或者也可以 git commit -m

“注释内容”，直接带注释提交

```
Administrator@WIN-F38TFIC4EUV MINGW64 /d/git/gittest (master)
$ git commit
[master (root-commit) b2d0cb6] create 01
1 file changed, 1 insertion(+)
create mode 100644 src/Test.java
```

提交完成后再查看状态

```
Administrator@WIN-F38TFIC4EUV MINGW64 /d/git/gittest (master)
$ git status
On branch master
nothing to commit, working tree clean
```

3.4 查看文件提交记录

1) 执行 `git log` 文件名 查看历史记录

```
Administrator@WIN-F38TFIC4EUV MINGW64 /d/git/gittest (master)
$ git log
commit b2d0cb64be476d2669b9f4fa03c7c78a529f9e31 (HEAD -> master)
Author: zhangsan <zhangsan@atguigu.com>
Date: Sun Dec 19 12:51:16 2021 +0800

    create 01
```

2) 通过 `git log --pretty=oneline` 文件名 简易信息查看

```
Administrator@WIN-F38TFIC4EUV MINGW64 /d/git/gittest (master)
$ git log --pretty=oneline
7ac488be40981827d8e9bd68afc2e70661d9dcc7 (HEAD -> master) update 03
372a00a984517d195241339f0e28de5f74933c88 update 02
bd93ad7648cbbd5c834e7e243e34bbdc99b5ece add Test.java
```

3.5 回退历史

1) `git reset --hard HEAD^` 回退到上一次提交. ^表示回退一步

```
Administrator@WIN-F38TFIC4EUV MINGW64 /d/git/gittest (master)
$ git reset --hard HEAD^
HEAD is now at 372a00a update 02
```

2) `git reset--hard HEAD~n` 回退 n 次操作

```
Administrator@WIN-F38TFIC4EUV MINGW64 /d/git/gittest (master)
$ git reset --hard HEAD~1
HEAD is now at bd93ad7 add Test.java
```

3.6 版本穿越

1) 进行查看历史记录的版本号，执行 `git reflog` 文件名

```
Administrator@WIN-F38TFIC4EUV MINGW64 /d/git/gittest (master)
$ git reflog
bd93ad7 (HEAD -> master) HEAD@{0}: reset: moving to HEAD~1
372a00a HEAD@{1}: reset: moving to HEAD^
7ac488b HEAD@{2}: commit: update 03
372a00a HEAD@{3}: commit: update 02
bd93ad7 (HEAD -> master) HEAD@{4}: commit (initial): add Test.java
```

2) 执行 `git reset --hard` 版本号

```
Administrator@WIN-F38TFIC4EUV MINGW64 /d/git/gittest (master)
$ git reset --hard 7ac488b
HEAD is now at 7ac488b update 03
```

3.7 还原文件

1) `git checkout` 文件名

如果文件修改，还未进行 `git add` 的情况下，可以通过还原回到修改之前的状态。如果已经提交了，只能使用 `git reset` 命令回退。

```
Administrator@WIN-F38TFIC4EUV MINGW64 /d/git/gittest (master)
$ cat src/Test.java
111111111111
222222222222
333333333333

Administrator@WIN-F38TFIC4EUV MINGW64 /d/git/gittest (master)
$ vim src/Test.java

Administrator@WIN-F38TFIC4EUV MINGW64 /d/git/gittest (master)
$ cat src/Test.java
111a111111aa111
222b222222bb222
333c333333cc333

Administrator@WIN-F38TFIC4EUV MINGW64 /d/git/gittest (master)
$ git checkout -- src/Test.java

Administrator@WIN-F38TFIC4EUV MINGW64 /d/git/gittest (master)
$ cat src/Test.java
111111111111
222222222222
333333333333
```

3.8 删除文件

1) 如果进行了一些文件的误提交，先将本地文件删除，再执行 `git add` 提交。

```
Administrator@WIN-F38TFIC4EUV MINGW64 /d/git/gittest (master)
$ touch src/girlfriend.jpg

Administrator@WIN-F38TFIC4EUV MINGW64 /d/git/gittest (master)
$ git add src

Administrator@WIN-F38TFIC4EUV MINGW64 /d/git/gittest (master)
$ git commit -m "new file"
[master 67df861] new file
1 file changed, 0 insertions(+), 0 deletions(-)
create mode 100644 src/girlfriend.jpg

Administrator@WIN-F38TFIC4EUV MINGW64 /d/git/gittest (master)
$ rm -rf src/girlfriend.jpg

Administrator@WIN-F38TFIC4EUV MINGW64 /d/git/gittest (master)
$ git status
On branch master
changes not staged for commit:
  (use "git add/rm <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        deleted:    src/girlfriend.jpg

no changes added to commit (use "git add" and/or "git commit -a")

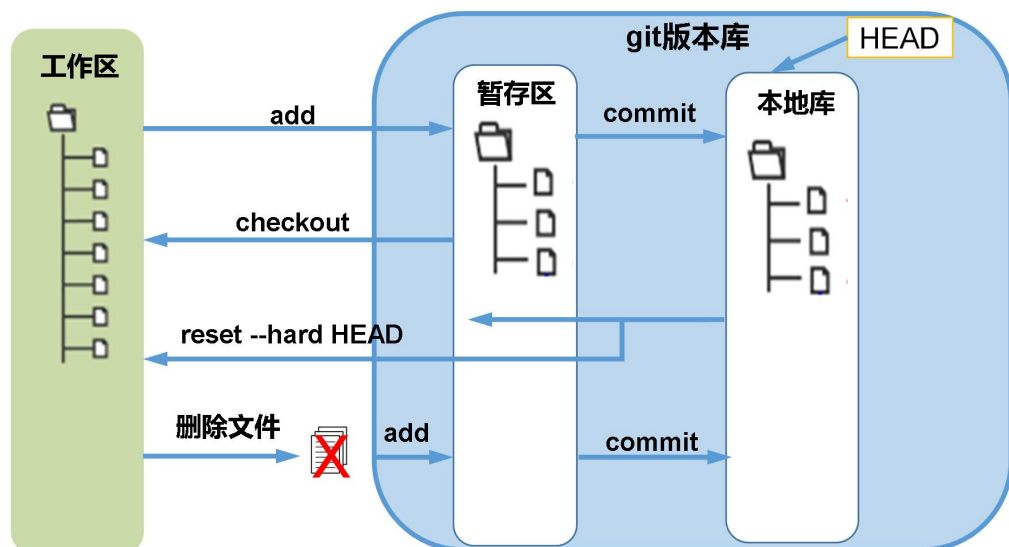
Administrator@WIN-F38TFIC4EUV MINGW64 /d/git/gittest (master)
$ git add src/girlfriend.jpg

Administrator@WIN-F38TFIC4EUV MINGW64 /d/git/gittest (master)
$ git commit -m "fix bug"
[master db683b0] fix bug
1 file changed, 0 insertions(+), 0 deletions(-)
delete mode 100644 src/girlfriend.jpg
```

2) 实际上, 只要提交后的文件, 在 git 中都有记录, 每个用户可以通过 `git reflog` 查看到对应的版本号, 还可以再回退到对应的版本。

3.9 三个区

Git实战操练—工作区+暂存区+本地库



让天下没有难学的技术

工作区:项目对应的本地磁盘目录

暂存区:一般存放在.git目录下的 index 文件.

本地库:项目对应的本地磁盘目录中的.git目录

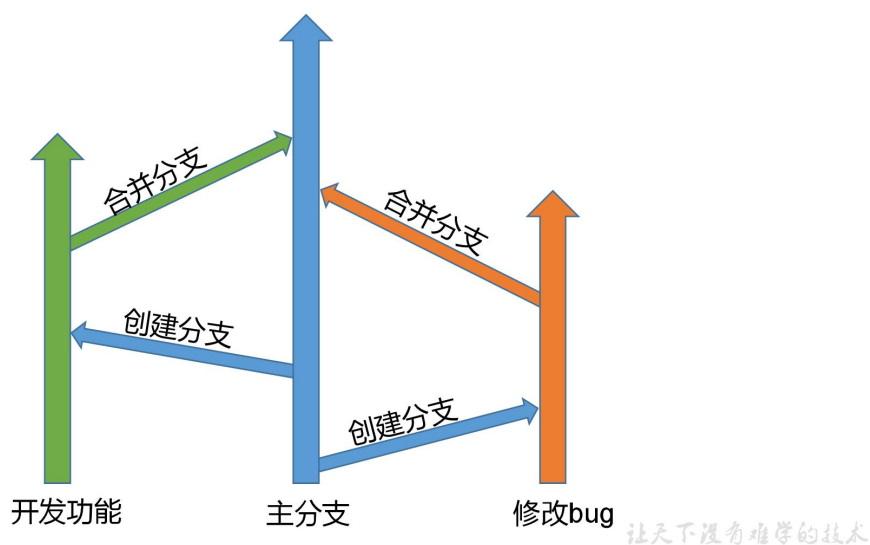
3.10 Git 分支

3.10.1 分支原理

系统上线了,但是产品经理又提了新的需求,评估一下工期要两个月,但是同时系统正在上线运行,时不时还要修改 bug, 如何管理几个版本?



Git分支



3.10.2 分支实战

1) 查看分支

```
git branch -v
```

```
Administrator@WIN-F38TFIC4EUV MINGW64 /d/git/gittest (master)
$ git branch -v
* master db683b0 fix bug
```

2) 创建分支

```
git branch <分支名>
```

```
Administrator@WIN-F38TFIC4EUV MINGW64 /d/git/gittest (master)
$ git branch dev

Administrator@WIN-F38TFIC4EUV MINGW64 /d/git/gittest (master)
$ git branch -v
dev db683b0 fix bug
* master db683b0 fix bug
```

3) 切换分支

git checkout <分支名>

一步完成: git checkout -b <分支名> 创建分支并切换到该分支

```
Administrator@WIN-F38TFIC4EUV MINGW64 /d/git/gittest (master)
$ git checkout dev
Switched to branch 'dev'

Administrator@WIN-F38TFIC4EUV MINGW64 /d/git/gittest (dev)
$ touch src/Hello.java

Administrator@WIN-F38TFIC4EUV MINGW64 /d/git/gittest (dev)
$ git add src/Hello.java

Administrator@WIN-F38TFIC4EUV MINGW64 /d/git/gittest (dev)
$ git commit -m "add Hello.java"
[dev e3874d0] add Hello.java
1 file changed, 0 insertions(+), 0 deletions(-)
create mode 100644 src/Hello.java

Administrator@WIN-F38TFIC4EUV MINGW64 /d/git/gittest (dev)
$ git branch -v
* dev      e3874d0 add Hello.java
master db683b0 fix bug

Administrator@WIN-F38TFIC4EUV MINGW64 /d/git/gittest (dev)
$ ll src
total 1
-rw-r--r-- 1 Administrator 197121 0 12月 19 14:42 Hello.java
-rw-r--r-- 1 Administrator 197121 39 12月 19 13:22 Test.java
```

```
Administrator@WIN-F38TFIC4EUV MINGW64 /d/git/gittest (dev)
$ git checkout master
Switched to branch 'master'

Administrator@WIN-F38TFIC4EUV MINGW64 /d/git/gittest (master)
$ ll src
total 1
-rw-r--r-- 1 Administrator 197121 39 12月 19 13:22 Test.java
```

4) 合并分支

先切换到主干(要合并到的分支) git checkout master

再合并分支 git merge <分支名>

```
Administrator@WIN-F38TFIC4EUV MINGW64 /d/git/gittest (dev)
$ git checkout master
Switched to branch 'master'

Administrator@WIN-F38TFIC4EUV MINGW64 /d/git/gittest (master)
$ git merge dev
Updating db683b0..e3874d0
Fast-forward
 src/Hello.java | 0
1 file changed, 0 insertions(+), 0 deletions(-)
create mode 100644 src/Hello.java

Administrator@WIN-F38TFIC4EUV MINGW64 /d/git/gittest (master)
$ git branch -v
dev      e3874d0 add Hello.java
* master e3874d0 add Hello.java
```

5) 删除分支

先切换到主干 git checkout master

```
git branch -D <分支名>
```

```
Administrator@WIN-F38TFIC4EUV MINGW64 /d/git/gittest (master)
$ git branch -D dev
Deleted branch dev (was e3874d0).

Administrator@WIN-F38TFIC4EUV MINGW64 /d/git/gittest (master)
$ git branch -v
* master e3874d0 add Hello.java
```

3.11 分支冲突

1) 如果两个分支同时修改一个文件，再进行合并分支时，就会出现冲突 MERGING.

```
Administrator@WIN-F38TFIC4EUV MINGW64 /d/git/gittest (master)
$ git checkout dev
Switched to branch 'dev'

Administrator@WIN-F38TFIC4EUV MINGW64 /d/git/gittest (dev)
$ vim src/Hello.java

Administrator@WIN-F38TFIC4EUV MINGW64 /d/git/gittest (dev)
$ git add src/Hello.java
warning: LF will be replaced by CRLF in src/Hello.java.
The file will have its original line endings in your working directory

Administrator@WIN-F38TFIC4EUV MINGW64 /d/git/gittest (dev)
$ git commit -m "add 1"
[dev e081d0d] add 1
1 file changed, 1 insertion(+)

Administrator@WIN-F38TFIC4EUV MINGW64 /d/git/gittest (master)
$ vim src/Hello.java

Administrator@WIN-F38TFIC4EUV MINGW64 /d/git/gittest (master)
$ git add src/Hello.java
warning: LF will be replaced by CRLF in src/Hello.java.
The file will have its original line endings in your working directory

Administrator@WIN-F38TFIC4EUV MINGW64 /d/git/gittest (master)
$ git commit -m "add 2"
[master 6d41226] add 2
1 file changed, 1 insertion(+)

Administrator@WIN-F38TFIC4EUV MINGW64 /d/git/gittest (master)
$ git branch -v
dev e081d0d add 1
* master 6d41226 add 2

Administrator@WIN-F38TFIC4EUV MINGW64 /d/git/gittest (master)
$ git merge dev
Auto-merging src/Hello.java
CONFLICT (content): Merge conflict in src/Hello.java
Automatic merge failed; fix conflicts and then commit the result.

Administrator@WIN-F38TFIC4EUV MINGW64 /d/git/gittest (master|MERGING)
$
```

2) 定位冲突

通过 git diff 定位冲突

```
Administrator@WIN-F38TFIC4EUV MINGW64 /d/git/gittest (master|MERGING)
$ git diff
diff --cc src/Hello.java
index 4c13393,cc583c4..0000000
--- a/src/Hello.java
+++ b/src/Hello.java
@@@ -1,1 -1,1 +1,5 @@@
++<<<<<< HEAD
++2222222222222222
++=====
++ 1111111111111111
++>>>>>> dev
```

3) 解决冲突

进入到冲突的文件, 进行内容的修改. 然后重新 git add 与 git commit

```
Administrator@WIN-F38TFIC4EUV MINGW64 /d/git/gittest (master|MERGING)
$ vim src/Hello.java

Administrator@WIN-F38TFIC4EUV MINGW64 /d/git/gittest (master|MERGING)
$ git add src/Hello.java

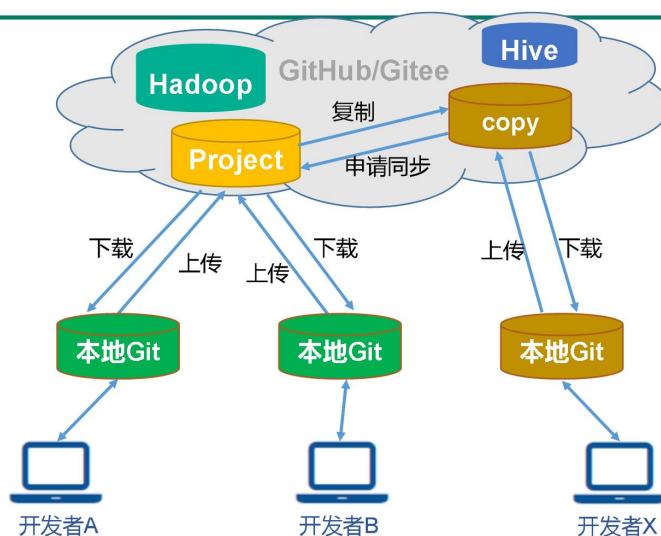
Administrator@WIN-F38TFIC4EUV MINGW64 /d/git/gittest (master|MERGING)
$ git commit -m "fix merge"
[master 218c794] fix merge

Administrator@WIN-F38TFIC4EUV MINGW64 /d/git/gittest (master)
$ git branch -v
dev    e081d0d add 1
* master 218c794 fix merge
```

第 4 章 GitHub 与 Gitee

4.1 是什么

GitHub (国外) 与 Gitee (国内) 是一个 Git 项目托管网站, 主要提供基于 Git 的版本托管服务。



让天下没有难学的技术

4.2 账号注册

1) GitHub 网址

<https://github.com/>

2) Gitee 网址

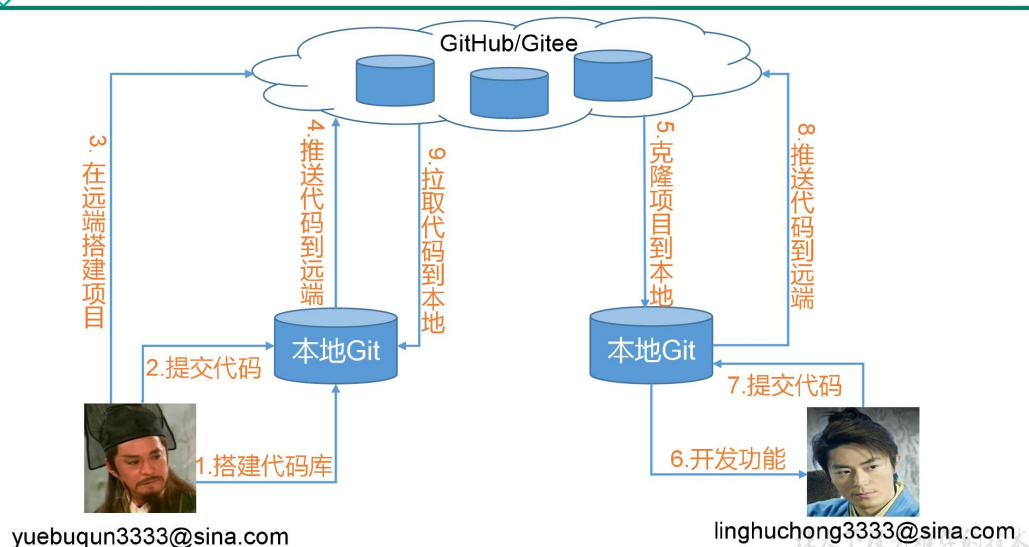
<https://gitee.com/>

4.3 实战操练



远端实操

尚硅谷



4.3.1 创建本地项目[岳不群]

1) 在本地版本库中新建 yuebuqun/jianfa 项目

```
Administrator@WIN-F38TFIC4EUV MINGW64 /d/git
$ pwd
/d/git

Administrator@WIN-F38TFIC4EUV MINGW64 /d/git
$ mkdir -p yuebuqun/jianfa

Administrator@WIN-F38TFIC4EUV MINGW64 /d/git
$ cd yuebuqun/jianfa/
```

2) 初始化版本库 git init, 设置签名

```
Administrator@WIN-F38TFIC4EUV MINGW64 /d/git/yuebuqun/jianfa (master)
$ git init
Initialized empty Git repository in D:/git/yuebuqun/jianfa/.git/

Administrator@WIN-F38TFIC4EUV MINGW64 /d/git/yuebuqun/jianfa (master)
$ git config user.name "岳不群"

Administrator@WIN-F38TFIC4EUV MINGW64 /d/git/yuebuqun/jianfa (master)
$ git config user.email "yuebuqun3333@sina.com"
```

3) 提交文件

```
Administrator@WIN-F38TFIC4EUV MINGW64 /d/git/yuebuqun/jianfa (master)
$ mkdir src

Administrator@WIN-F38TFIC4EUV MINGW64 /d/git/yuebuqun/jianfa (master)
$ cd src

Administrator@WIN-F38TFIC4EUV MINGW64 /d/git/yuebuqun/jianfa/src (master)
$ touch bixiejianfa.java

Administrator@WIN-F38TFIC4EUV MINGW64 /d/git/yuebuqun/jianfa/src (master)
$ vim bixiejianfa.java
Administrator@WIN-F38TFIC4EUV MINGW64 /d/git/yuebuqun/jianfa/src (master)
$ cd ..

Administrator@WIN-F38TFIC4EUV MINGW64 /d/git/yuebuqun/jianfa (master)
$ git status
On branch master

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
  src/

Administrator@WIN-F38TFIC4EUV MINGW64 /d/git/yuebuqun/jianfa (master)
$ git add src/
warning: LF will be replaced by CRLF in src/bixiejianfa.java.
The file will have its original line endings in your working directory

Administrator@WIN-F38TFIC4EUV MINGW64 /d/git/yuebuqun/jianfa (master)
$ git status
On branch master

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
    new file:   src/bixiejianfa.java

Administrator@WIN-F38TFIC4EUV MINGW64 /d/git/yuebuqun/jianfa (master)
$ git commit -m "jianfa init"
[master (root-commit) f9124a4] jianfa init
1 file changed, 1 insertion(+)
create mode 100644 src/bixiejianfa.java
```

4.3.2 在 Gitee 中创建 Project

1) 创建仓库



创建仓库时默认是私有，可以在创建好后在设置中将仓库改为开源。



创建完成后，可以查看到工程地址，后面会用到。



4.3.3 增加远程地址

1) git remote add <远端代号> <远端地址>。

<远端代号> 是指远程链接的代号，一般直接用 origin 作代号，也可以自定义。

<远端地址> 默认远程链接的 url

```
Administrator@WIN-F38TFIC4EUV MINGW64 /d/git/yuebuqun/jianfa (master)
$ git remote add origin https://gitee.com/yuebuqun3333/jianfa.git
```

4.3.4 推送到远端库

1) git push <远端代号> <本地分支名称>。

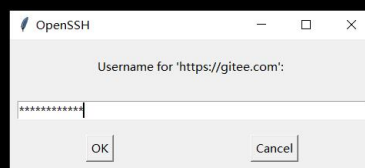
<远端代号> 是指远程链接的代号。

<分支名称> 是指要提交的分支名字，比如 master。

例：git push origin master

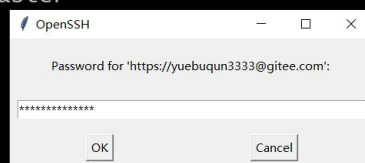
2) 输入用户名

```
Administrator@WIN-F38TFIC4EUV MINGW64 /d/git/yuebuqun/jianfa (master)
$ git push origin master
```



3) 输入密码

```
Administrator@WIN-F38TFIC4EUV MINGW64 /d/git/yuebuqun/jianfa (master)
$ git push origin master
```



4) 推送成功后显示

```
Administrator@WIN-F38TFIC4EUV MINGW64 /d/git/yuebuqun/jianfa (master)
$ git push origin master
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Writing objects: 100% (4/4), 285 bytes | 95.00 KiB/s, done.
Total 4 (delta 0), reused 0 (delta 0)
remote: Powered by GITEE.COM [GNK-6.2]
To https://gitee.com/yuebuqun3333/jianfa.git
 * [new branch]      master -> master
```

5) 在 Gitee 中查看



4.3.5 从 GitHub 上克隆一个项目[令狐冲]

1) 创建 linghuchong 目录

```
Administrator@WIN-F38TFIC4EUV MINGW64 /d/git
$ pwd
/d/git

Administrator@WIN-F38TFIC4EUV MINGW64 /d/git
$ mkdir linghuchong
```

2) 从 Gitee 中克隆项目到本地

`git clone <远端地址> <新项目目录名>。`

<远端地址> 是指远程链接的地址。

<项目目录名> 是指为克隆的项目在本地新建的目录名称，可以不填，默认是 Gitee 的项目名。

命令执行完后，会自动为这个远端地址建一个名为 origin 的代号。

```
Administrator@WIN-F38TFIC4EUV MINGW64 /d/git
$ cd linghuchong/

Administrator@WIN-F38TFIC4EUV MINGW64 /d/git/linghuchong
$ git clone https://gitee.com/yuebuqun3333/jianfa.git
Cloning into 'jianfa'...
remote: Enumerating objects: 4, done.
remote: Counting objects: 100% (4/4), done.
remote: Total 4 (delta 0), reused 0 (delta 0), pack-reused 0
Unpacking objects: 100% (4/4), done.
```

3) 设置当前版本库的用户角色

```
Administrator@WIN-F38TFIC4EUV MINGW64 /d/git/linghuchong/jianfa (master)
$ git config user.name "令狐冲"

Administrator@WIN-F38TFIC4EUV MINGW64 /d/git/linghuchong/jianfa (master)
$ git config user.email "linghuchong3333@sina.com"
```

4) 修改克隆下来的代码


```
Administrator@WIN-F38TFIC4EUV MINGW64 /d/git/linghuchong/jianfa (master)
$ cd src/

Administrator@WIN-F38TFIC4EUV MINGW64 /d/git/linghuchong/jianfa/src (master)
$ vim bixiejianfa.java

Administrator@WIN-F38TFIC4EUV MINGW64 /d/git/linghuchong/jianfa/src (master)
$ git add bixiejianfa.java

Administrator@WIN-F38TFIC4EUV MINGW64 /d/git/linghuchong/jianfa/src (master)
$ git commit -m "啥玩意"
[master e6207ec] 啥玩意
1 file changed, 1 insertion(+)
```

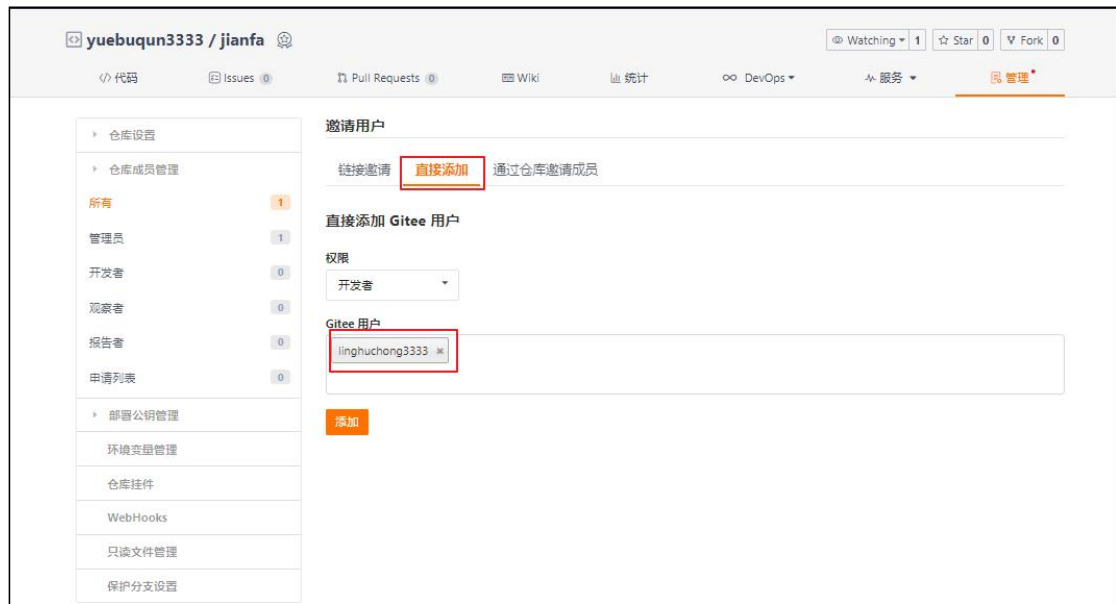
4.3.6 将新的更新提交到 Gitee[令狐冲]

1) 失败: 对于 Gitee 来说, 不允许别的用户更新内容, 必须将别的用户添加到仓库成员中

```
Administrator@WIN-F38TFIC4EUV MINGW64 /d/git/linghuchong/jianfa/src (master)
$ git push origin master
remote: [session-e6403bbc] Access denied
fatal: unable to access 'https://gitee.com/yuebuqun3333/jianfa.git/': The request
ed URL returned error: 403
```

2) 邀请成员

The top screenshot shows the Gitee repository page for 'yuebuqun3333 / jianfa'. The '仓库管理' (Repository Management) button is highlighted in a red box. The bottom screenshot shows the '仓库成员管理' (Repository Member Management) page. The '添加仓库成员' (Add Repository Member) button is highlighted in a red box. The left sidebar shows the '仓库成员管理' section expanded, with '开发者' (Developer) highlighted.



3) 同意邀请





4) 同意以后，在 Gitee 中可以看到项目信息



5) 接下来重新进行提交操作

```
Administrator@WIN-F38TFIC4EUV MINGW64 /d/git/linghuchong/jianfa/src (master)
$ git push origin master
Enumerating objects: 7, done.
Counting objects: 100% (7/7), done.
Writing objects: 100% (4/4), 338 bytes | 338.00 KiB/s, done.
Total 4 (delta 0), reused 0 (delta 0)
remote: Powered by GITEE.COM [GNK-6.2]
To https://gitee.com/yuebuqun3333/jianfa.git
f9124a4..e6207ec master -> master
```

6) 提交成功后，岳不群可看到令狐冲对项目的更新。



4.3.7 从 Gitee 中更新项目到本地库【岳不群】

1) `git pull <远端代号> <远端分支名>`。

<远端代号> 是指远程链接的代号。

<远端分支名>是指远端的分支名称，如 `master`。

例 `git pull origin master`

```
Administrator@WIN-F38TFIC4EUV MINGW64 /d/git/yuebuqun/jianfa (master)
$ git pull origin master
remote: Enumerating objects: 7, done.
remote: Counting objects: 100% (7/7), done.
remote: Total 4 (delta 0), reused 0 (delta 0), pack-reused 0
Unpacking objects: 100% (4/4), done.
From https://gitee.com/yuebuqun3333/jianfa
* branch          master      -> FETCH_HEAD
  f9124a4..e6207ec master      -> origin/master
Updating f9124a4..e6207ec
Fast-forward
 src/bixiejianfa.java | 1 +
 1 file changed, 1 insertion(+)
```

2) 查看更新的文件

```
Administrator@WIN-F38TFIC4EUV MINGW64 /d/git/yuebuqun/jianfa (master)
$ cat src/bixiejianfa.java
欲练此功....
啥玩意???
```

4.3.8 协作冲突

多个用户修改项目中的同一个文件提交到 GitHub 会出现冲突问题


```
Administrator@WIN-F38TFIC4EUV MINGW64 /d/git/linghuchong/jianfa (master)
$ git push origin master
To https://gitee.com/yuebuqun3333/jianfa.git
! [rejected]        master -> master (fetch first)
error: failed to push some refs to 'https://gitee.com/yuebuqun3333/jianfa.git'
hint: Updates were rejected because the remote contains work that you do
hint: not have locally. This is usually caused by another repository pushing
hint: to the same ref. You may want to first integrate the remote changes
hint: (e.g., 'git pull ...') before pushing again.
hint: See the 'Note about fast-forwards' in 'git push --help' for details.
```

5) 令狐冲更新项目到本地

```
Administrator@WIN-F38TFIC4EUV MINGW64 /d/git/linghuchong/jianfa (master)
$ git pull origin master
remote: Enumerating objects: 7, done.
remote: Counting objects: 100% (7/7), done.
remote: Total 4 (delta 0), reused 0 (delta 0), pack-reused 0
Unpacking objects: 100% (4/4), done.
From https://gitee.com/yuebuqun3333/jianfa
* branch        master      -> FETCH_HEAD
   e6207ec..3a8c954 master    -> origin/master
Auto-merging src/bixiejianfa.java
CONFLICT (content): Merge conflict in src/bixiejianfa.java
Automatic merge failed; fix conflicts and then commit the result.

Administrator@WIN-F38TFIC4EUV MINGW64 /d/git/linghuchong/jianfa (master|MERGING)
$ |
```

6) 定位冲突

```
Administrator@WIN-F38TFIC4EUV MINGW64 /d/git/linghuchong/jianfa (master|MERGING)
$ git diff
diff --cc src/bixiejianfa.java
index d17e606,c13af99..0000000
--- a/src/bixiejianfa.java
+++ b/src/bixiejianfa.java
@@@ -1,3 -1,3 +1,7 @@@
    欲练此功...
    啥玩意???
++<<<<<<< HEAD
+ 问问小师弟...
++=====
+ 问问别人
++>>>>>>> 3a8c954de44bb0e501a269a312ba736effc65bb6
```

7) 解决冲突

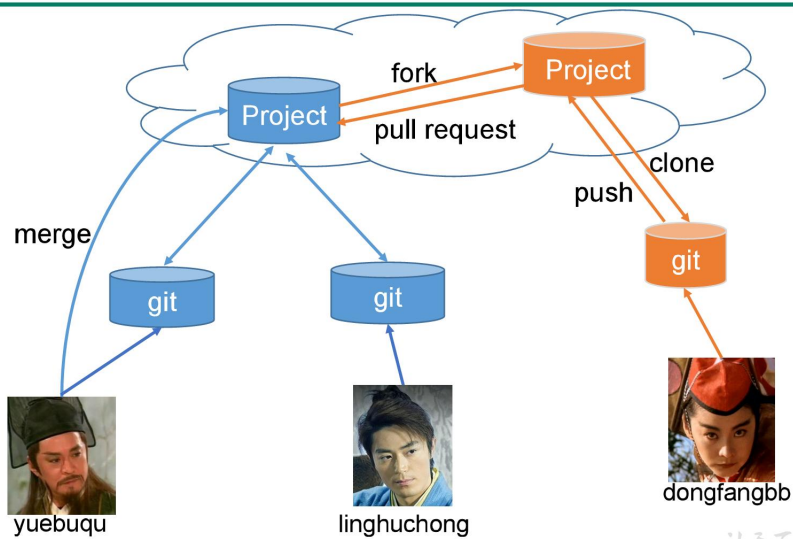
```
Administrator@WIN-F38TFIC4EUV MINGW64 /d/git/linghuchong/jianfa (master|MERGING)
$ vim src/bixiejianfa.java

Administrator@WIN-F38TFIC4EUV MINGW64 /d/git/linghuchong/jianfa (master|MERGING)
$ git add src/bixiejianfa.java

Administrator@WIN-F38TFIC4EUV MINGW64 /d/git/linghuchong/jianfa (master|MERGING)
$ git commit -m "fix merging"
[master 67c797a] fix merging

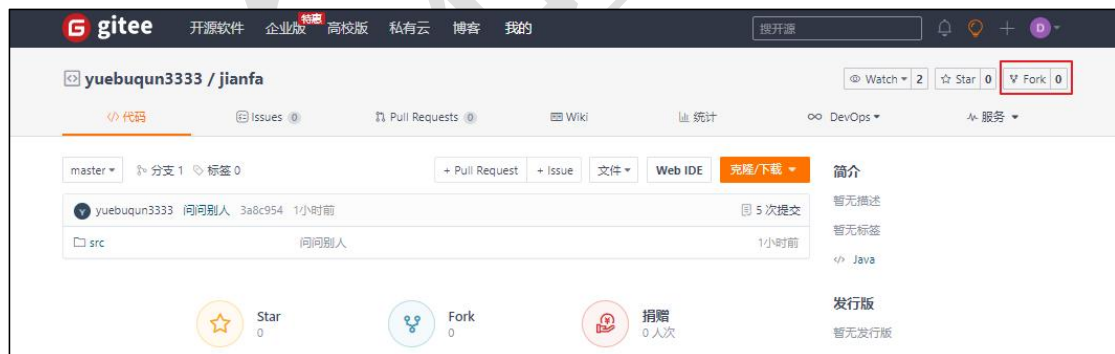
Administrator@WIN-F38TFIC4EUV MINGW64 /d/git/linghuchong/jianfa (master)
$ git push origin master
Enumerating objects: 11, done.
Counting objects: 100% (11/11), done.
Delta compression using up to 8 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (5/5), 560 bytes | 280.00 KiB/s, done.
Total 5 (delta 0), reused 0 (delta 0)
remote: Powered by GITEE.COM [GNK-6.2]
To https://gitee.com/yuebuqun3333/jianfa.git
   3a8c954..67c797a master -> master
```

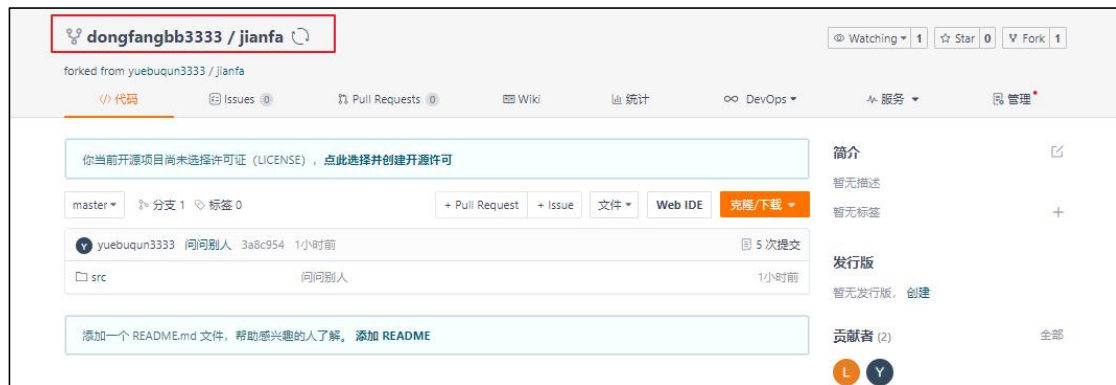
4.3.9 Git Fork



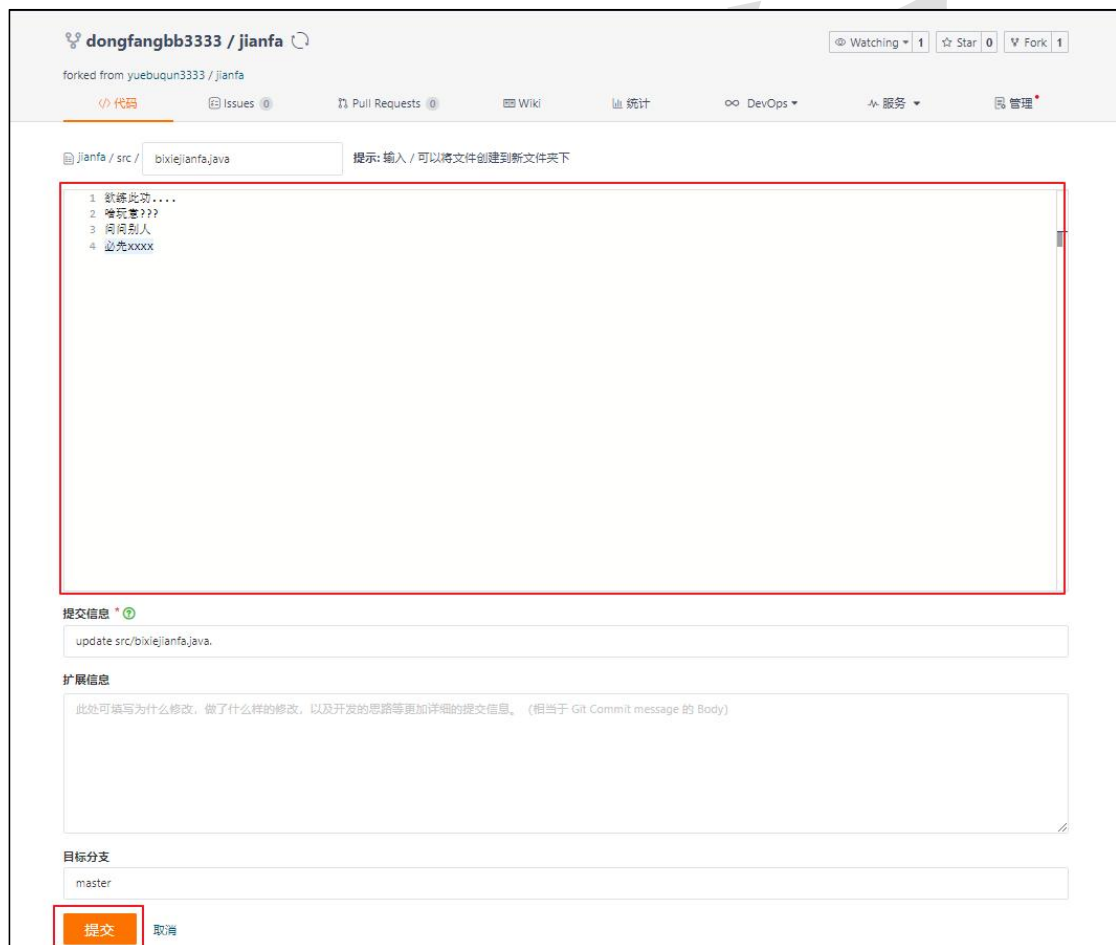
让天下没有难学的技术

1) 东方不败 fork 项目

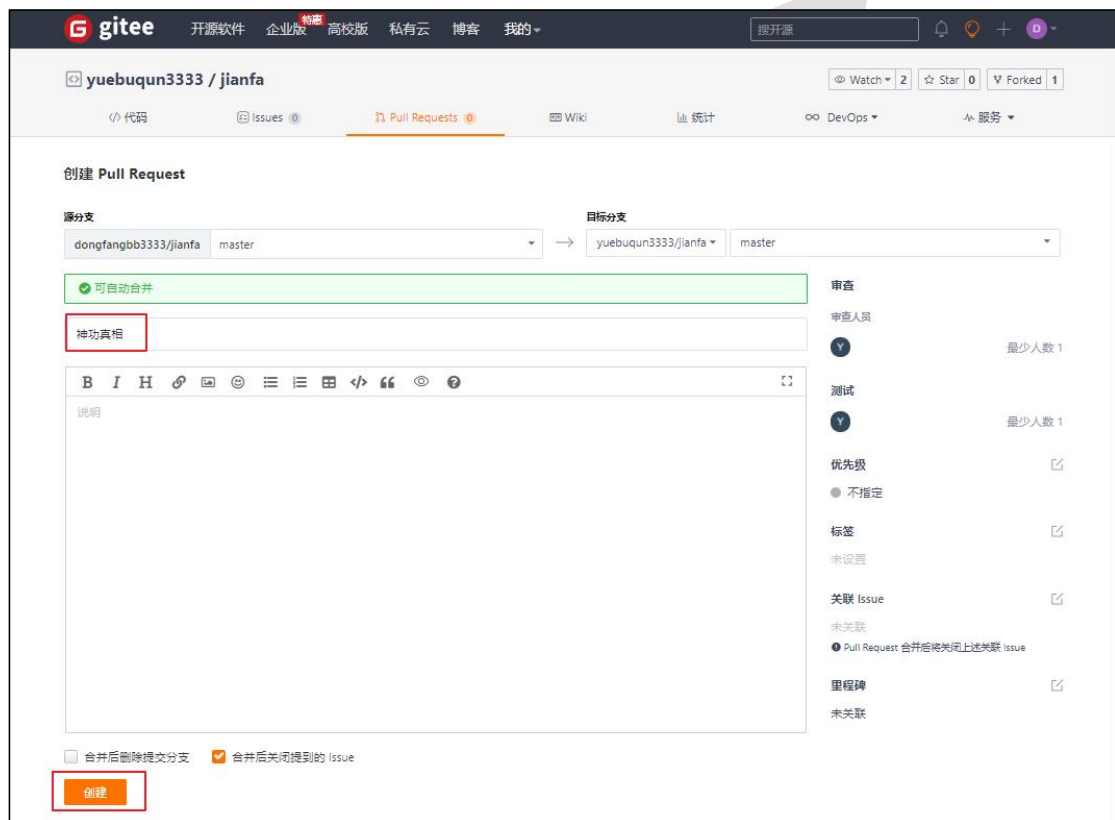
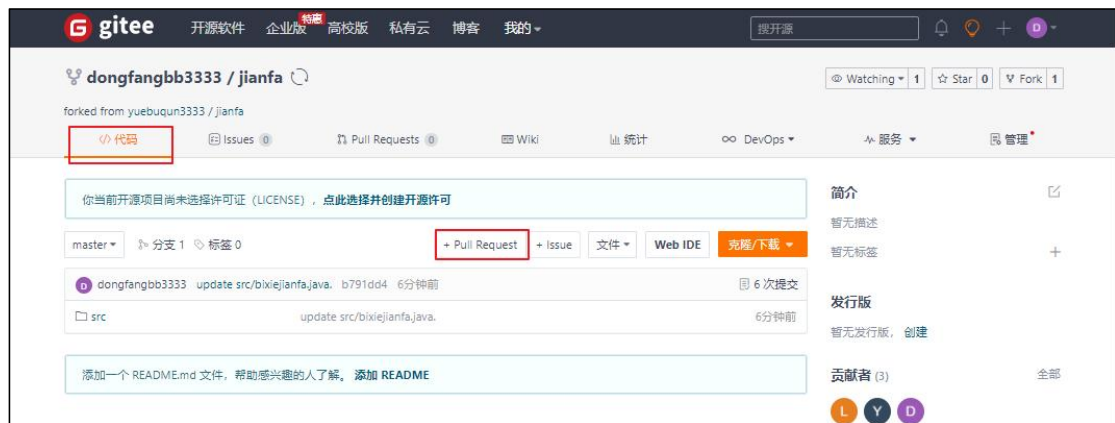




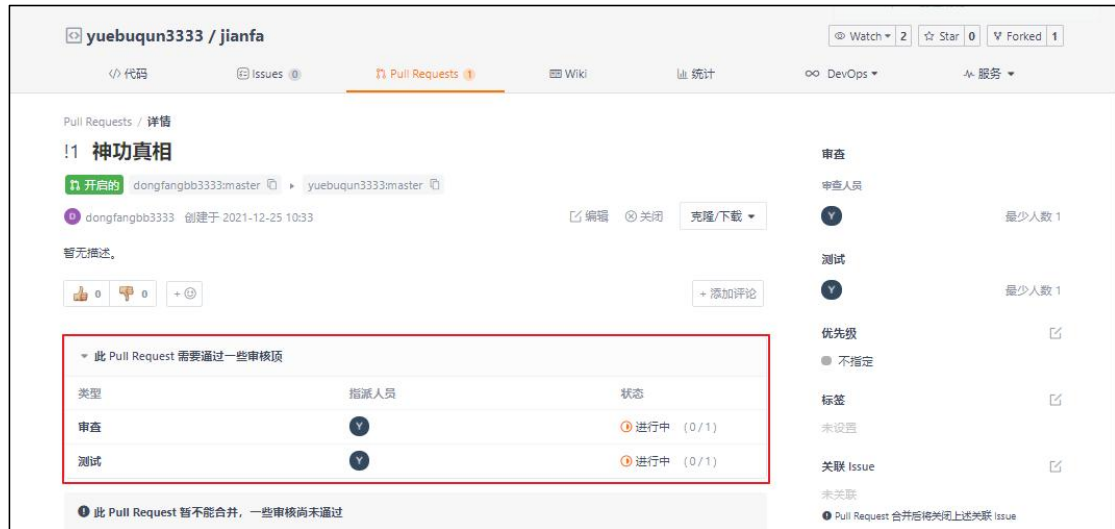
2) 查看项目并进行修改, 正常应该将项目更新到本地,也可直接在 gitee 中进行修改



3) 发送合并请求



4) 等待审查

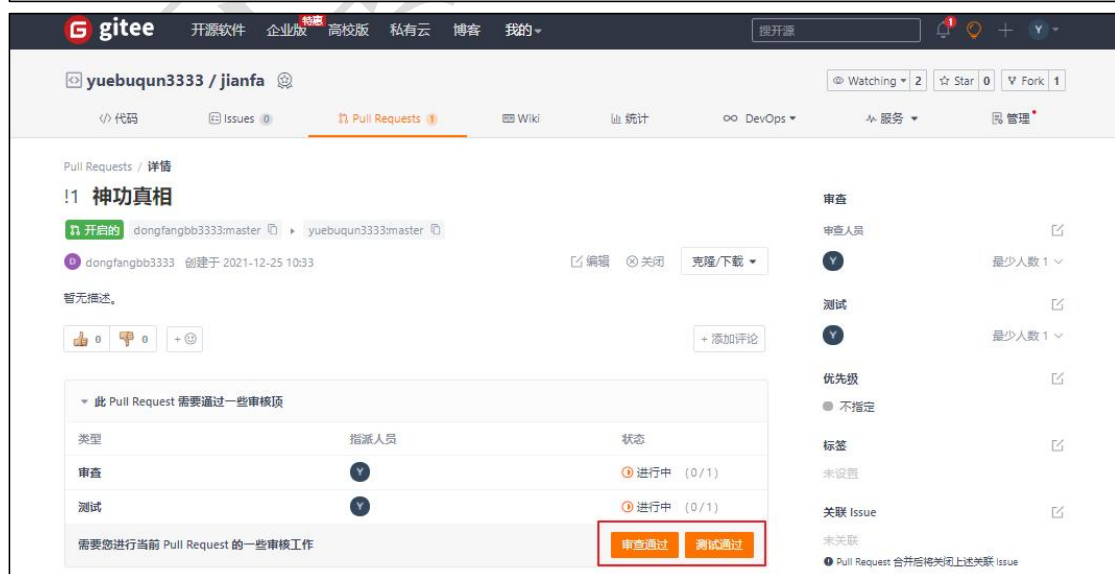


The screenshot shows a Pull Request titled "神功真相" (Divine Power Truth) on the Gitee platform. The PR is created by "dongfangbb3333" and is currently in the "进行中" (In Progress) state. The review status is shown as "0/1" for both "审查" (Review) and "测试" (Test). A red box highlights the review status section, indicating that the PR needs to pass some reviews before it can be merged.

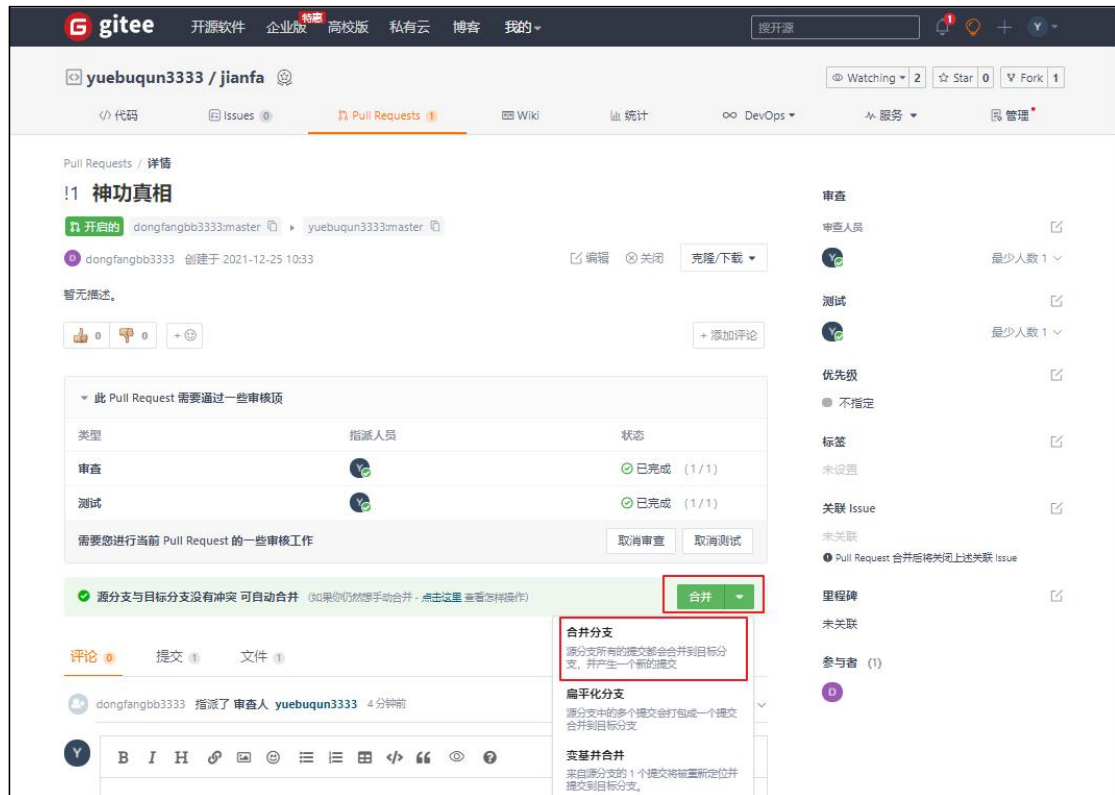
5) 岳不群审查



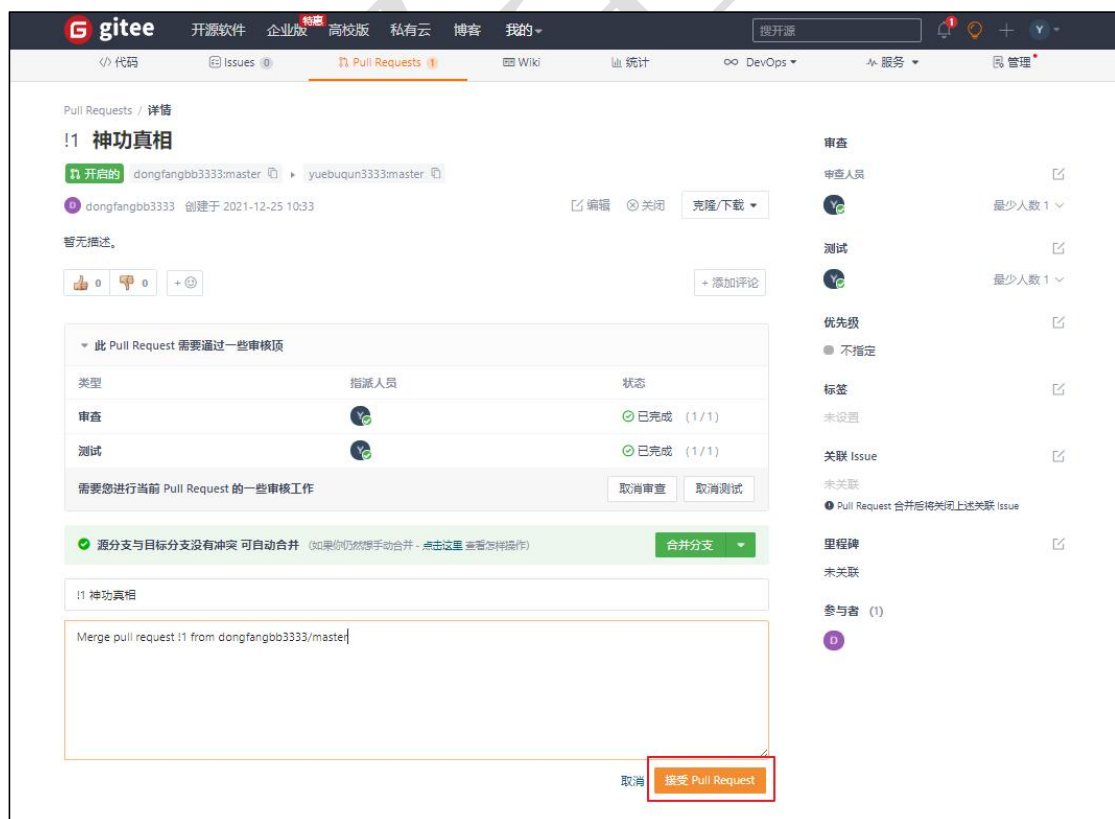
The screenshot shows the Gitee Pull Requests page for the user "yuebuqun3333". The page displays a list of Pull Requests, with the "神功真相" (Divine Power Truth) PR highlighted. The PR is currently in the "进行中" (In Progress) state. The review status is shown as "0/1" for both "审查" (Review) and "测试" (Test). A red box highlights the "审查" (Review) status, indicating that the PR needs to pass the review before it can be merged.



The screenshot shows the Gitee Pull Request page for the "神功真相" (Divine Power Truth) PR. The PR is created by "dongfangbb3333" and is currently in the "进行中" (In Progress) state. The review status is shown as "0/1" for both "审查" (Review) and "测试" (Test). A red box highlights the review status section, indicating that the PR needs to pass some reviews before it can be merged. At the bottom of the page, there are buttons for "审查通过" (Review Passed) and "测试通过" (Test Passed).



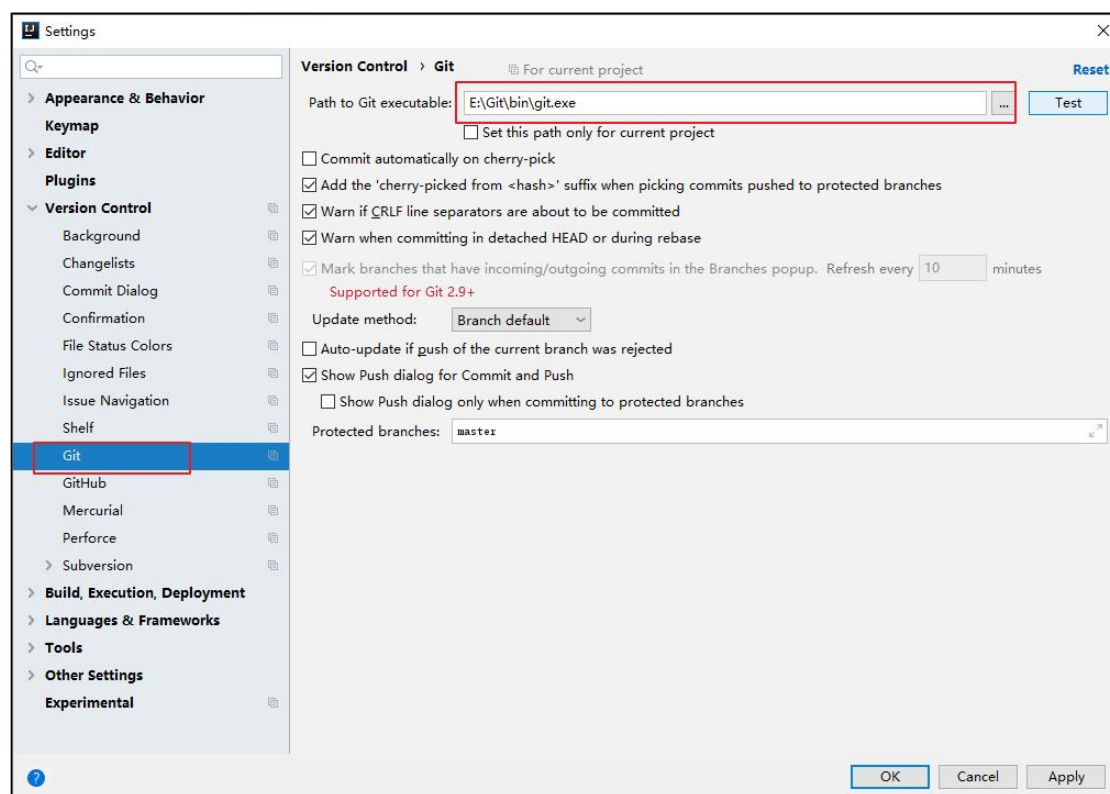
The screenshot shows the Gitee web interface for a pull request. The title is '!1 神功真相'. The pull request is from 'dongfangbb3333:master' to 'yuebuqun3333:master'. It was created on 2021-12-25 10:33. The status is '已开启' (Open). The pull request has 0 comments and 0 files. The '合并' (Merge) button is highlighted with a red box. The '合并分支' (Merge branch) dropdown menu is open, showing options: '合并分支' (Merge branch), '扁平化分支' (Flatten branch), and '变基合并' (Rebase merge). The '合并分支' option is selected.



The screenshot shows the Gitee web interface for a pull request. The title is '!1 神功真相'. The pull request is from 'dongfangbb3333:master' to 'yuebuqun3333:master'. It was created on 2021-12-25 10:33. The status is '已开启' (Open). The pull request has 0 comments and 0 files. The '接受 Pull Request' (Accept Pull Request) button is highlighted with a red box. The 'Merge pull request !1 from dongfangbb3333/master' text is visible in the merge box.

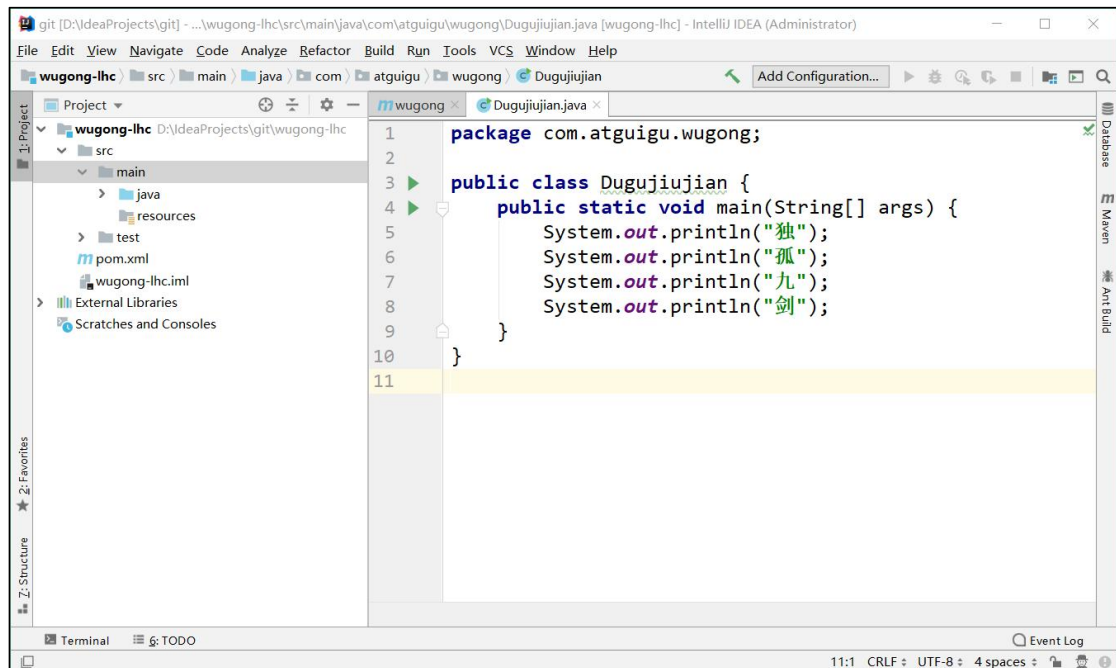
第 5 章 IDEA Git 操作

5.1 配置 Git

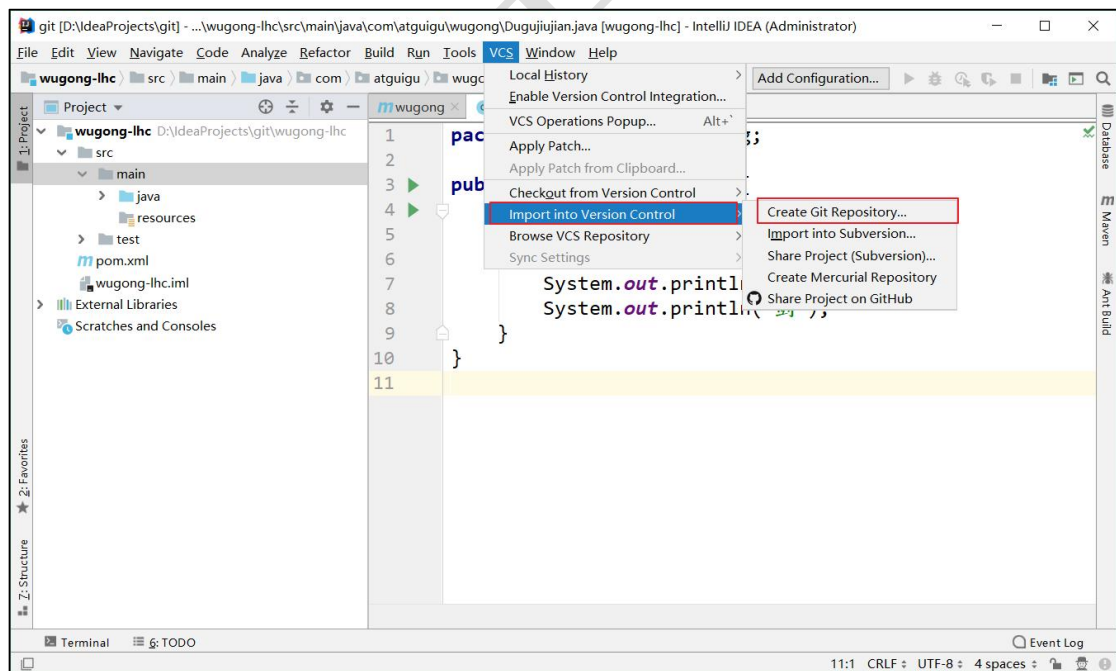


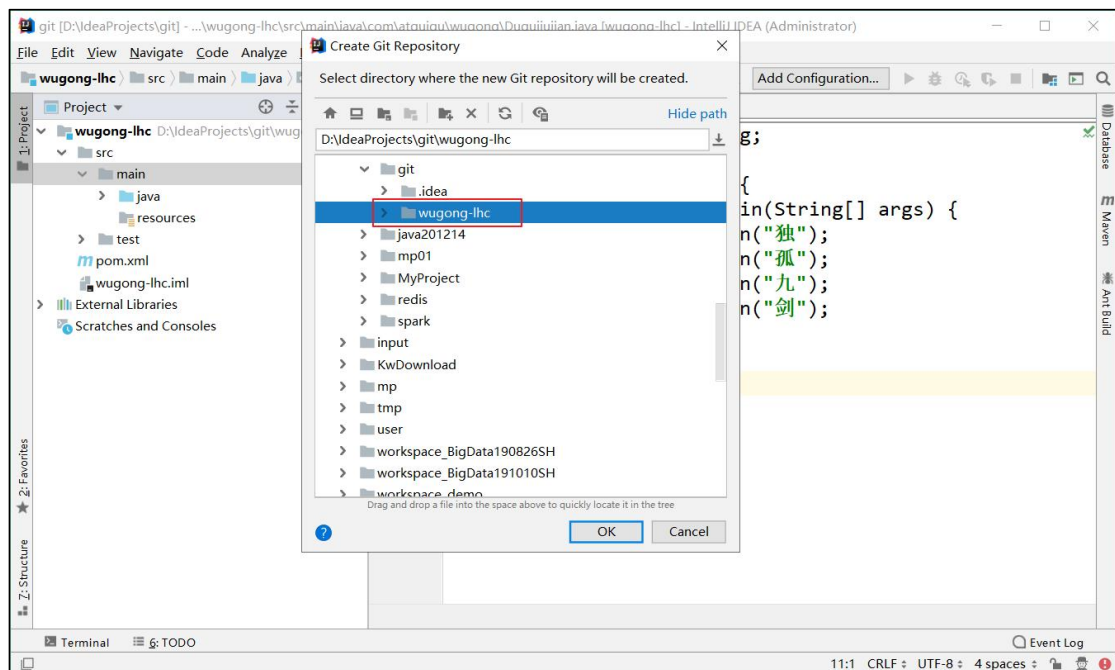
5.2 提交操作

1) 创建本地项目

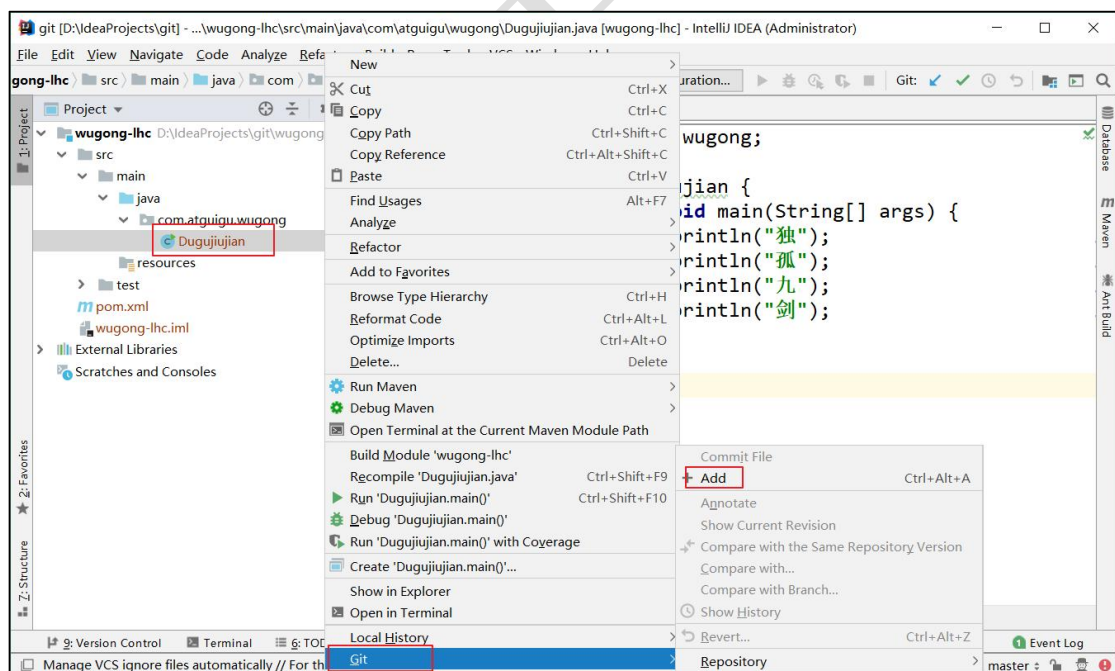


2) 初始化本地库

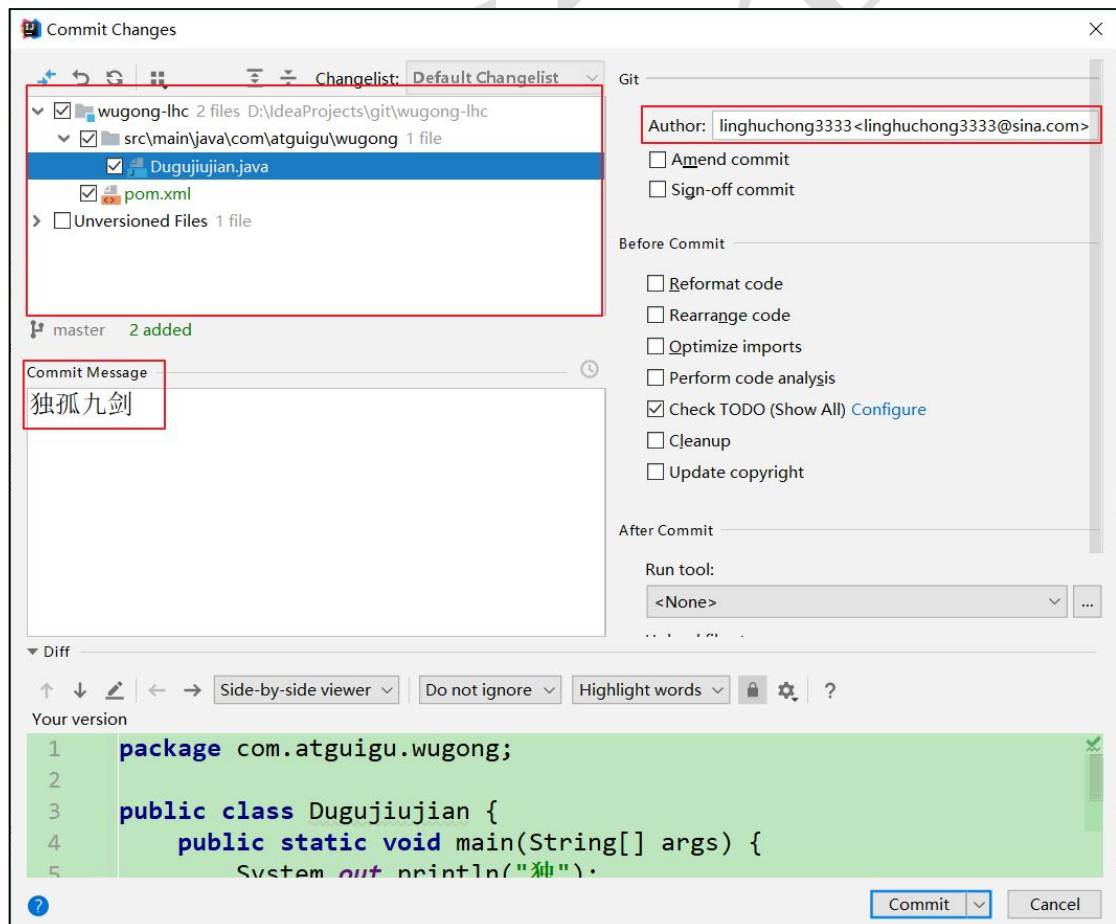
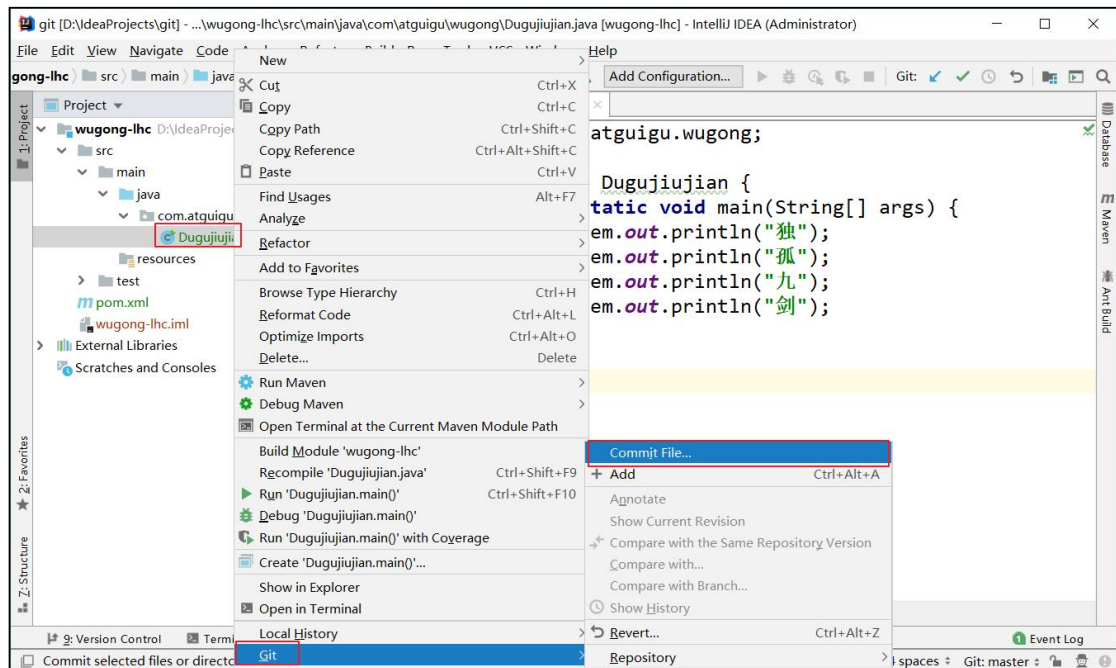




3) 加入本地库

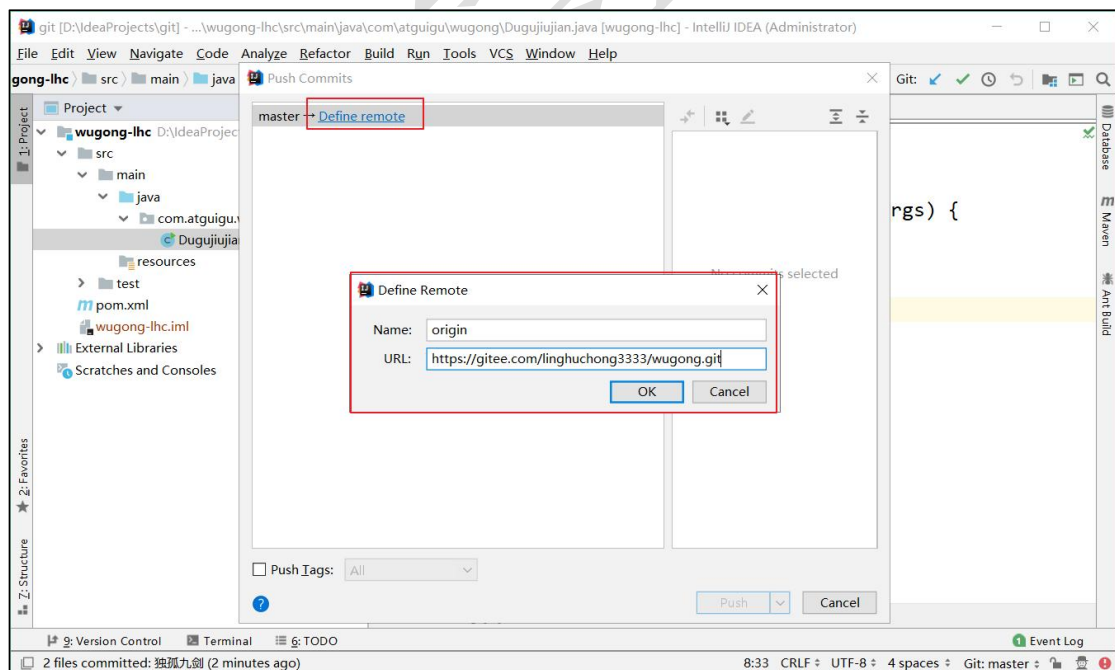
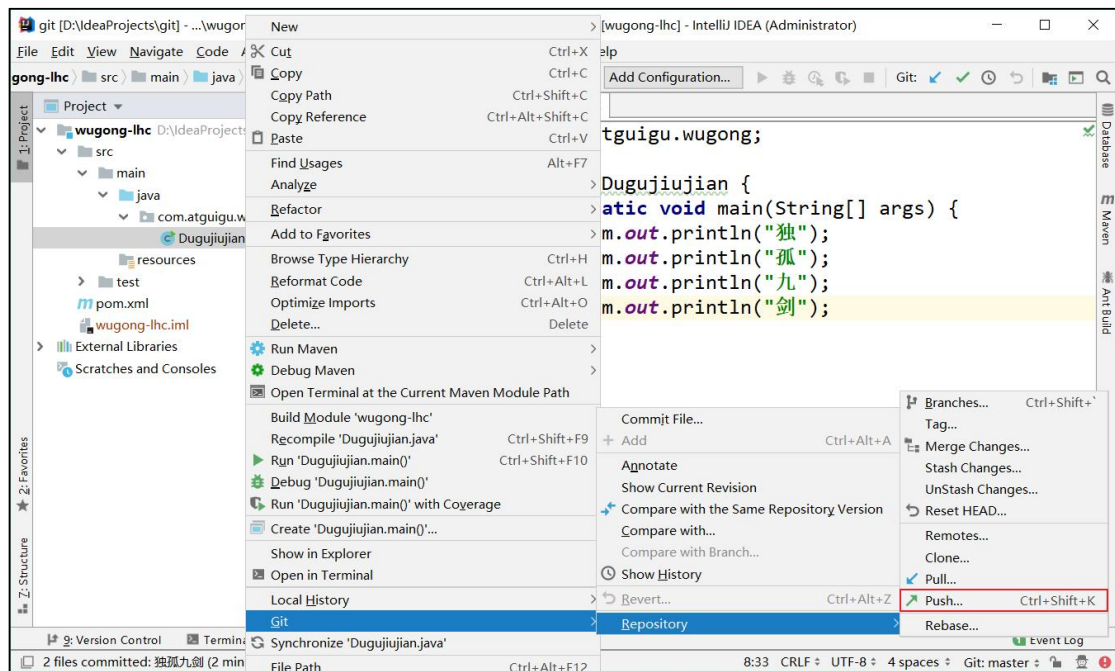


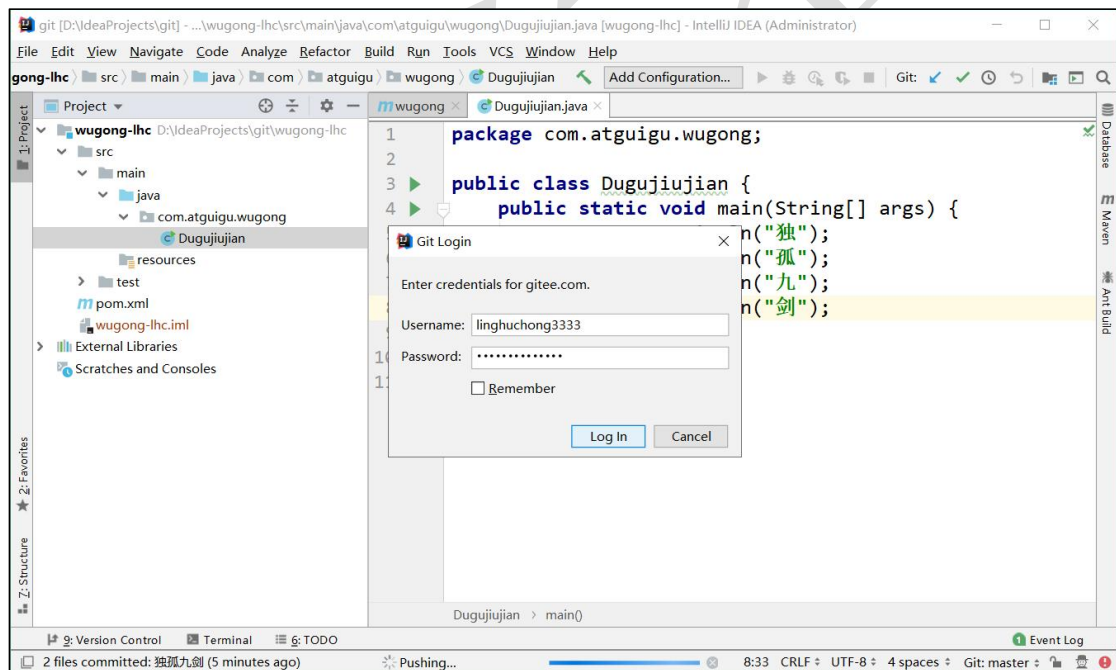
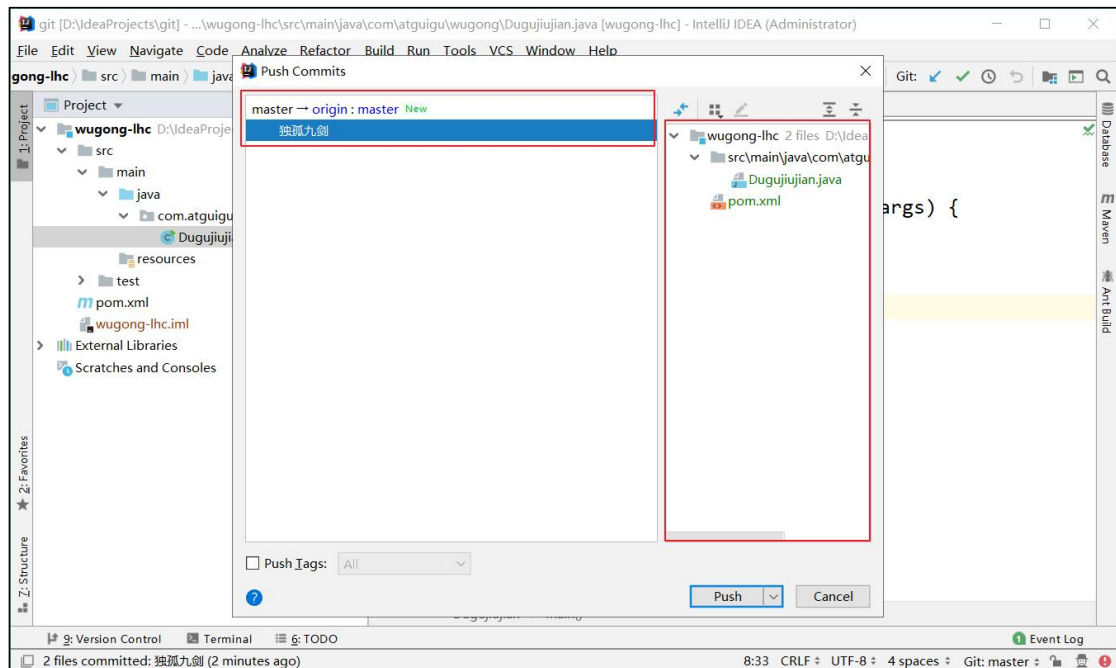
4) 提交



5) 在 gitee 中创建仓库

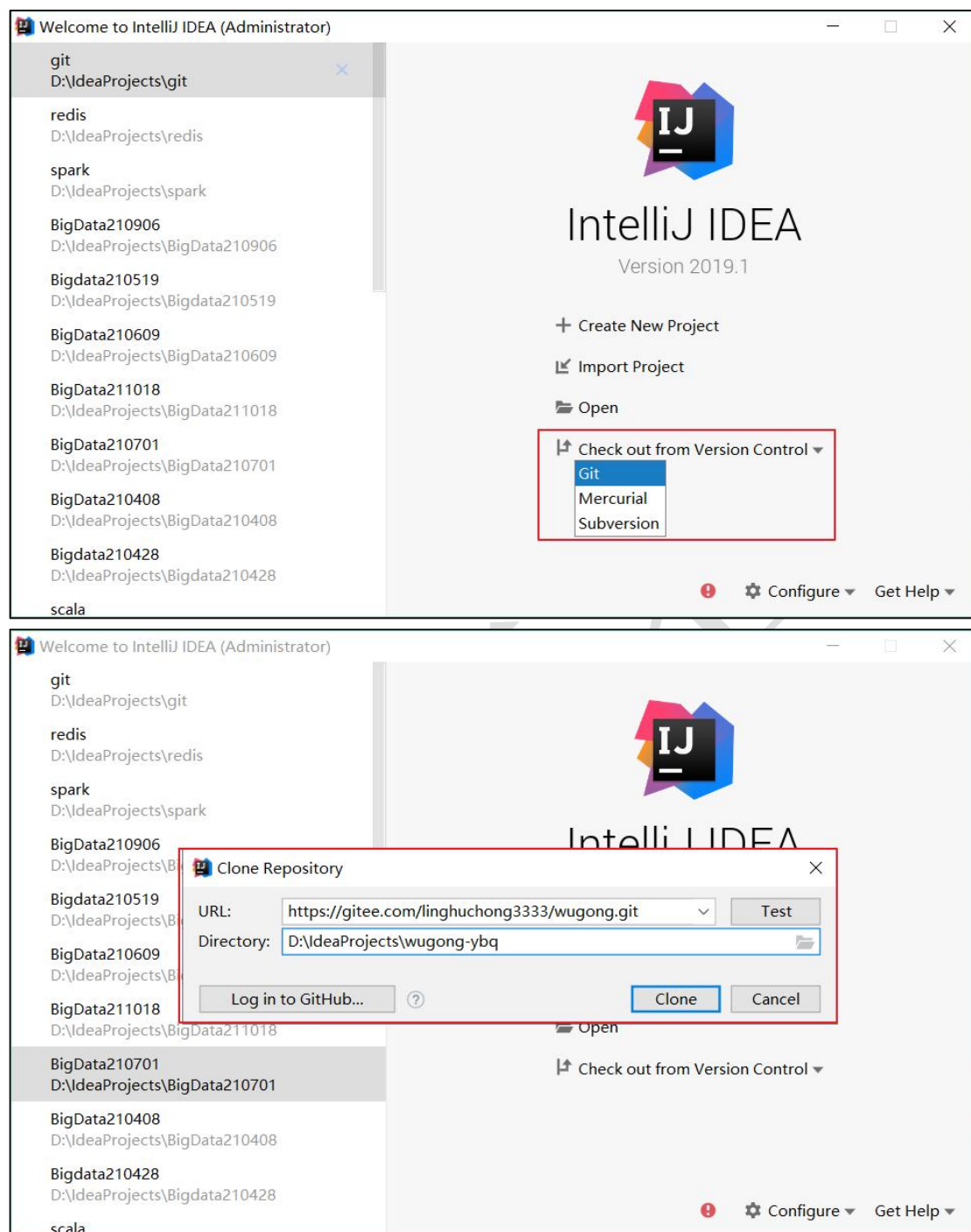
6) push 到 gitee

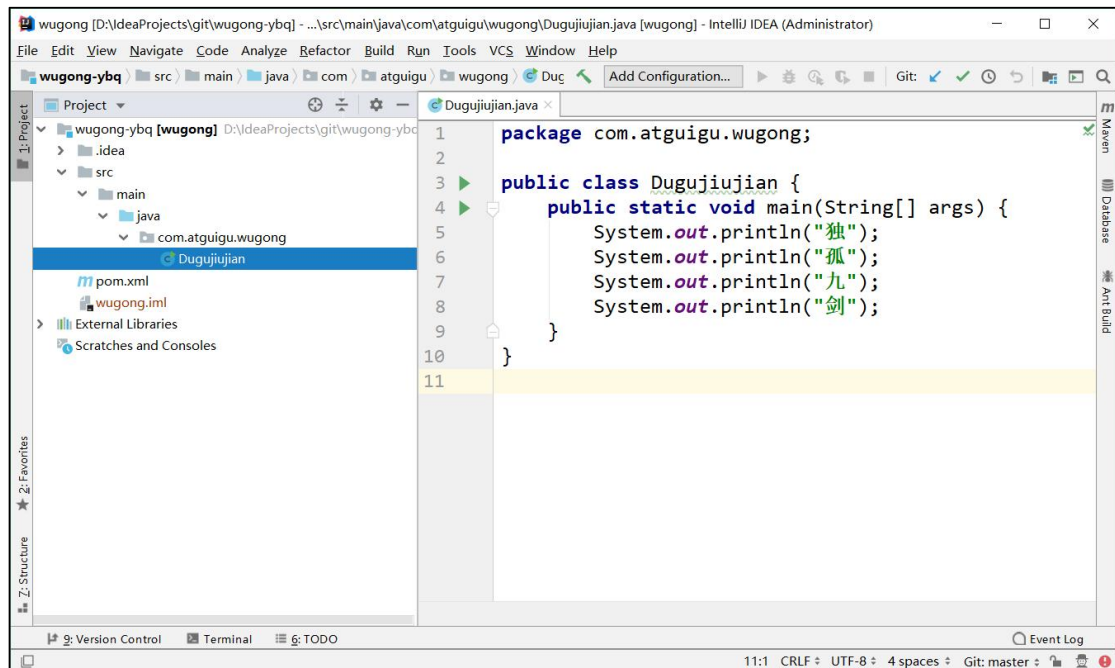




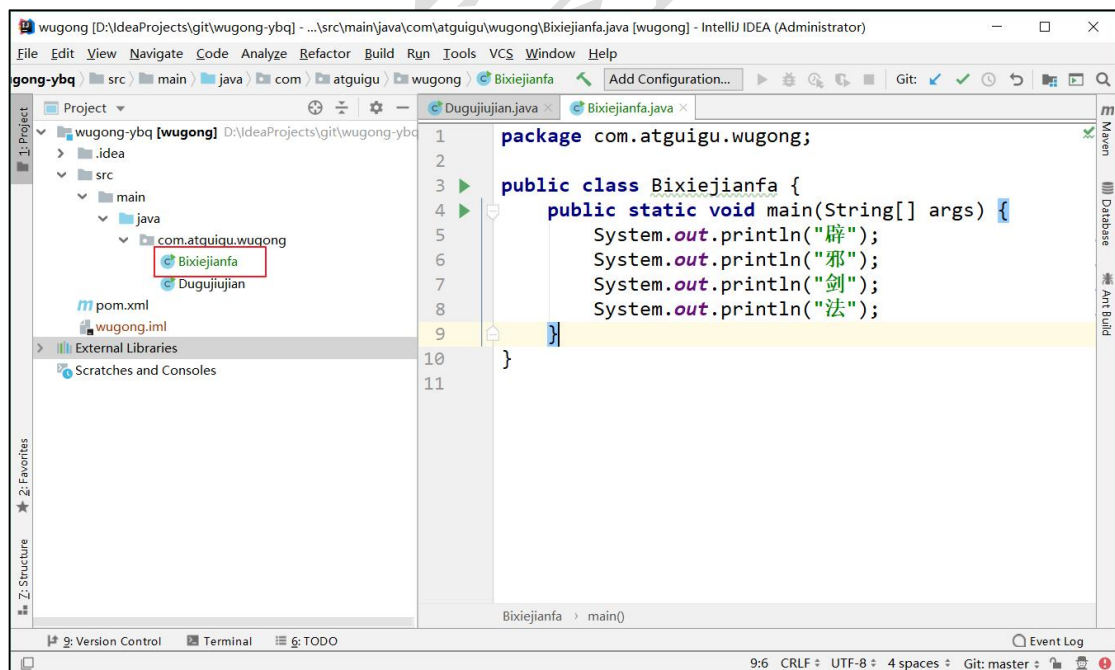
5.3 克隆到本地

1) 从 gitee 中 checkout

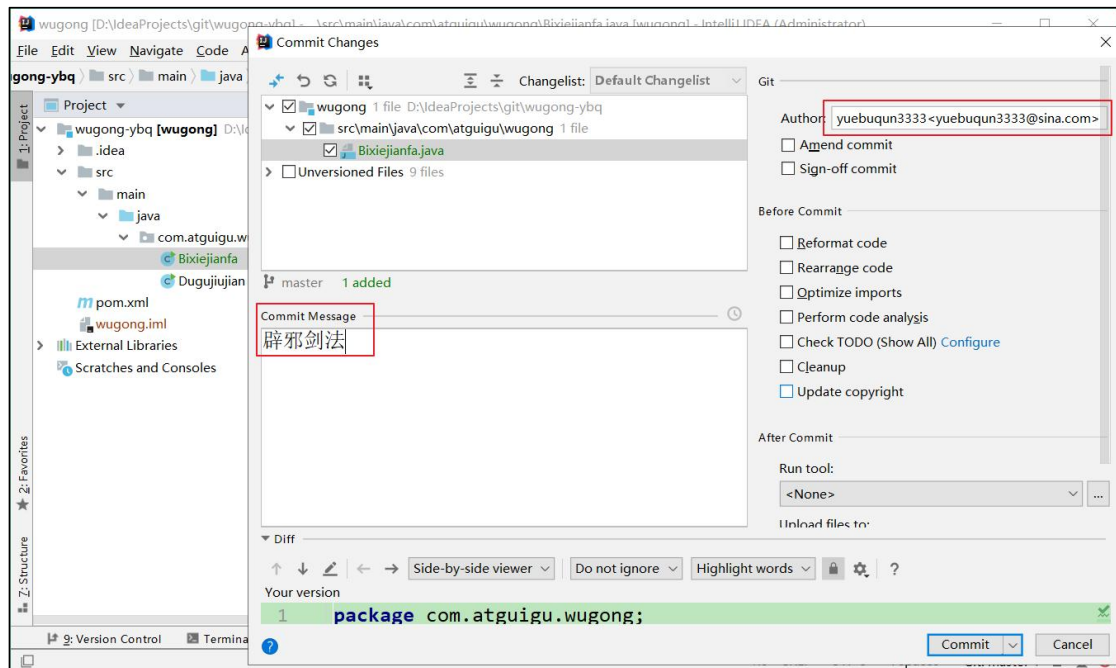




2) 添加代码

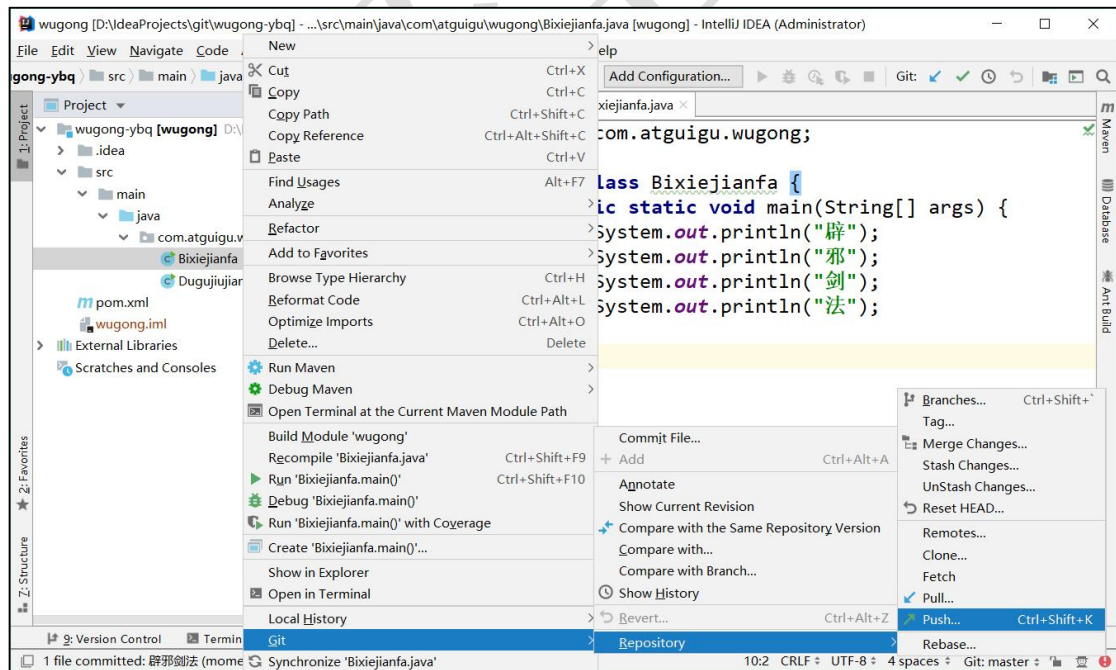


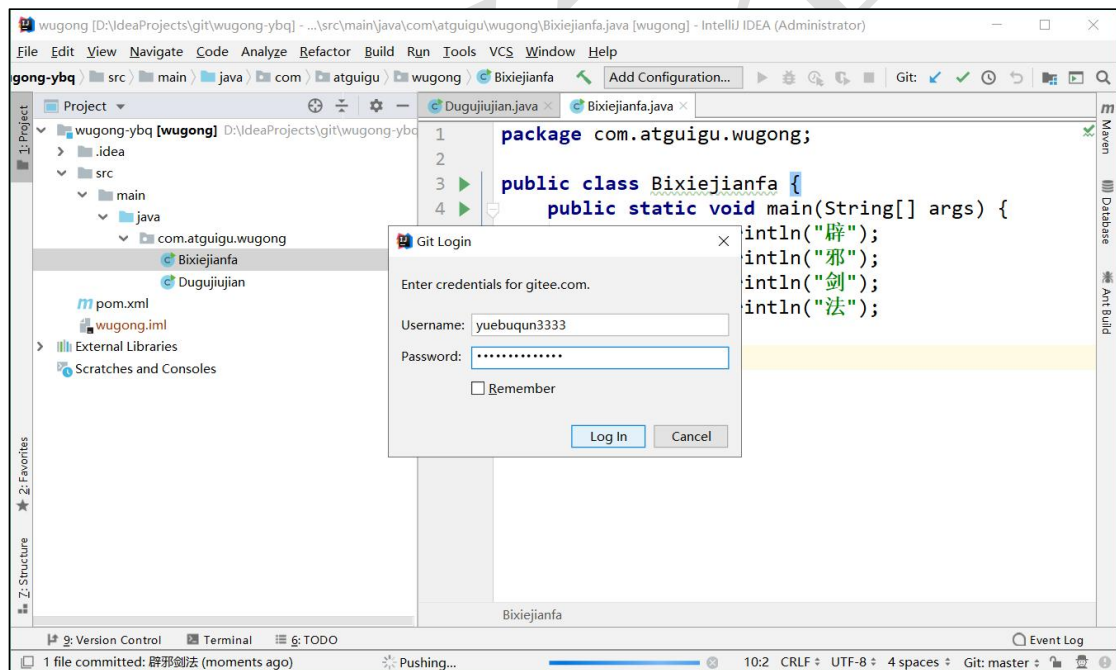
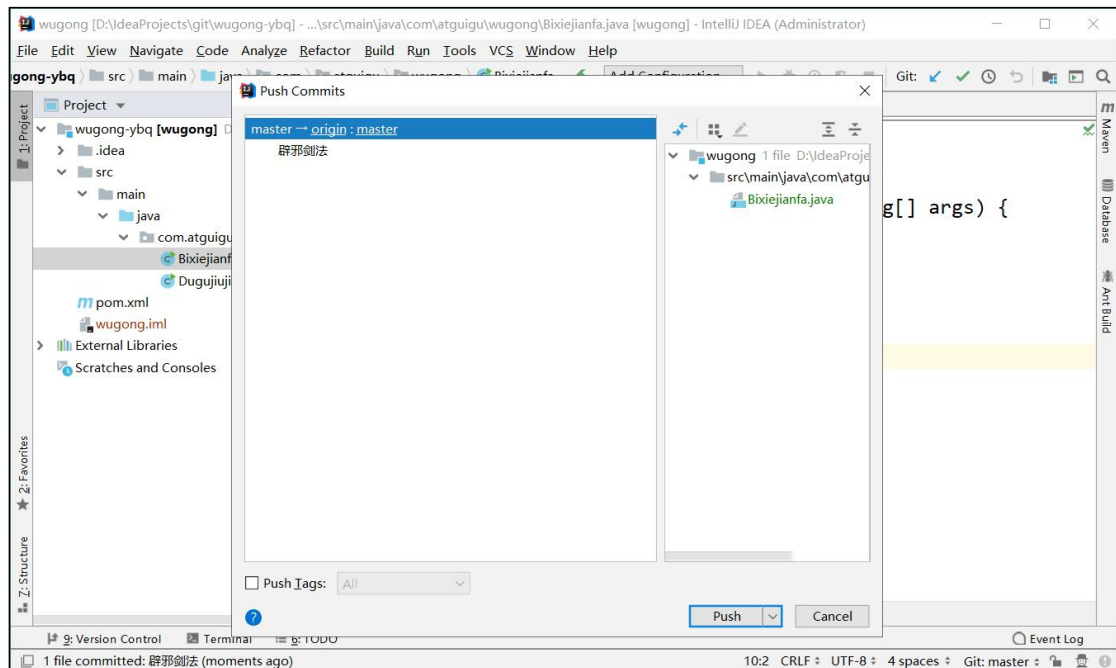
3) Commit



4) 加入到合作伙伴

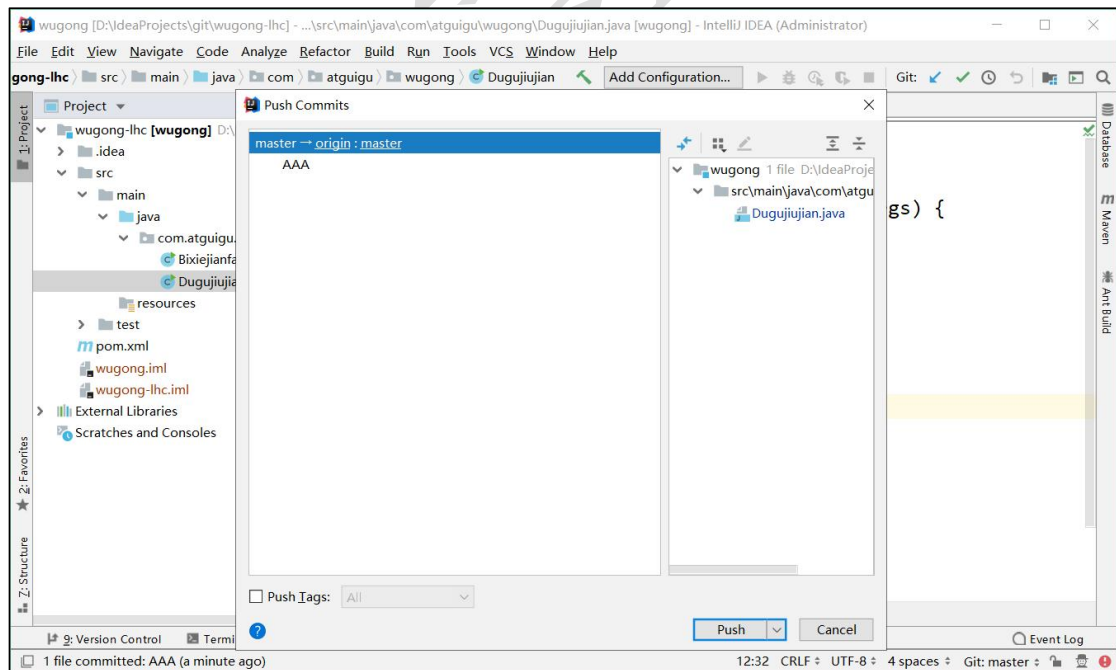
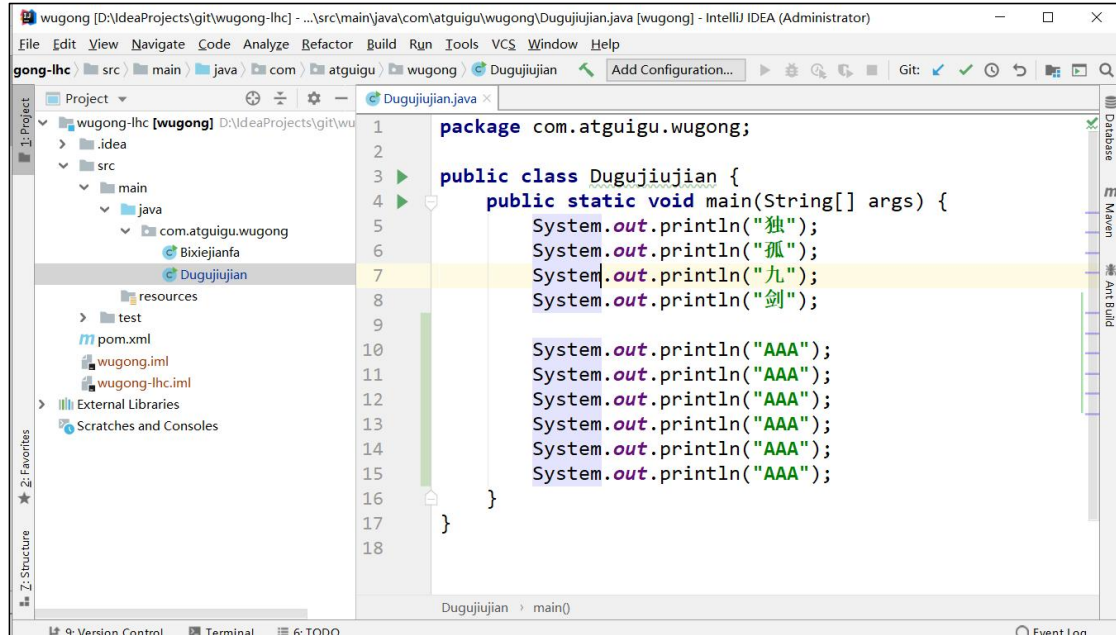
5) Push



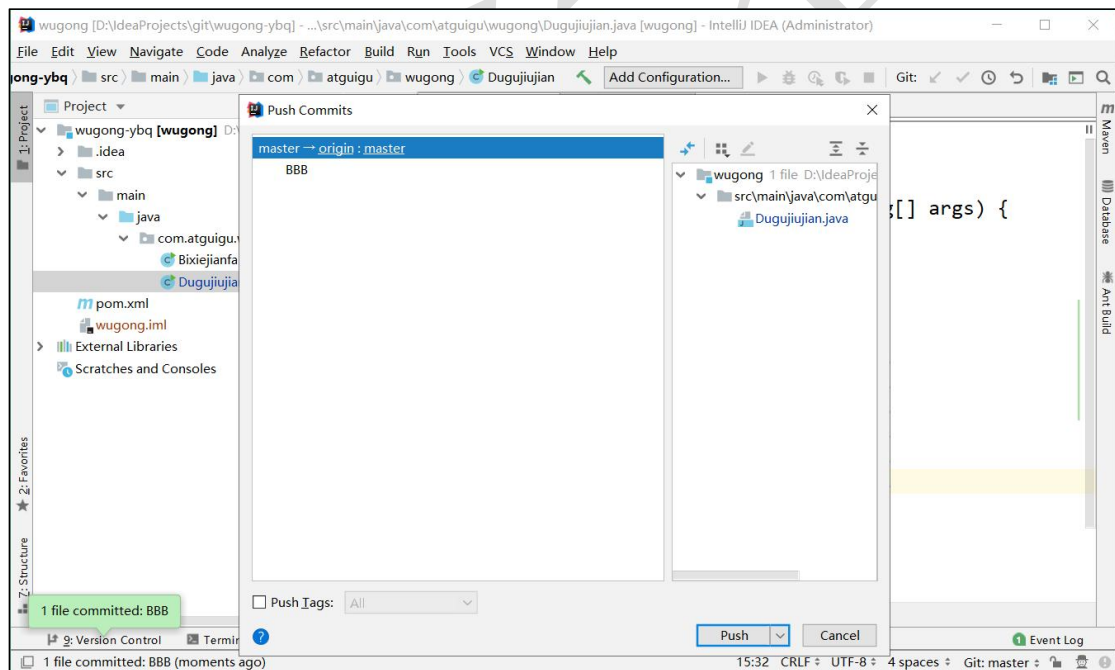
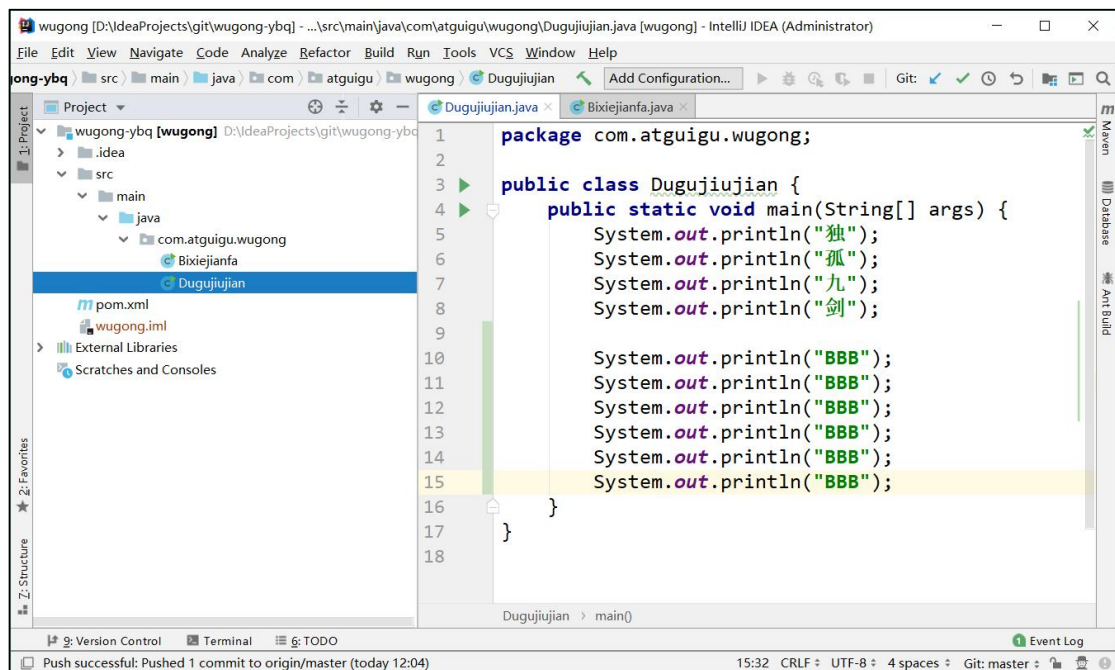


5.4 冲突解决

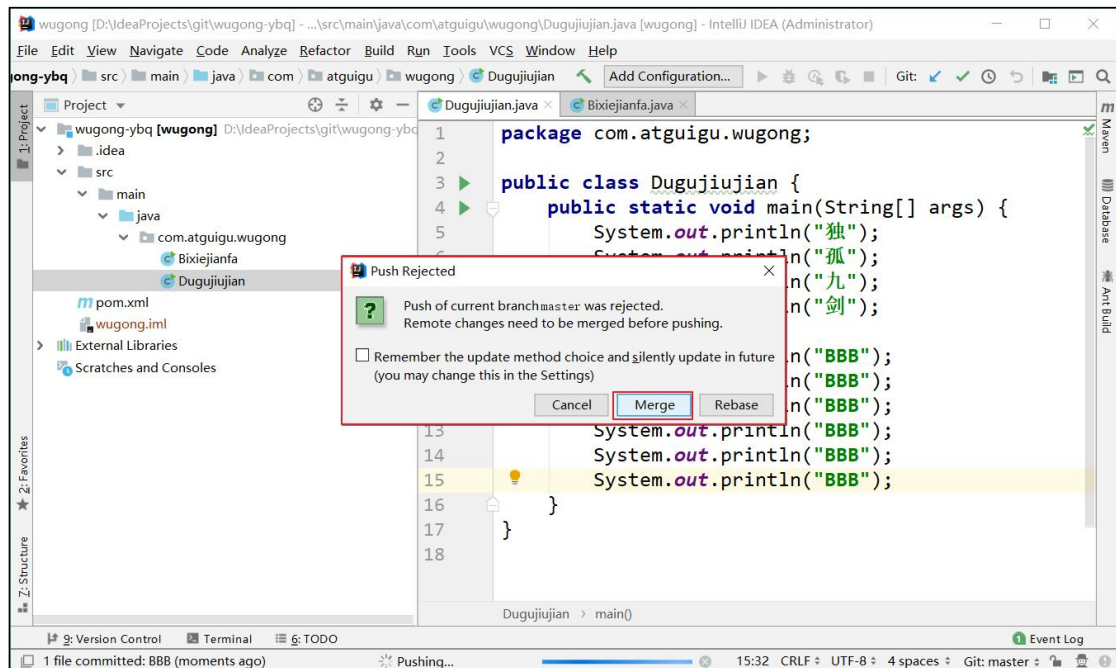
1) 令狐冲修改代码正常进行 commit push



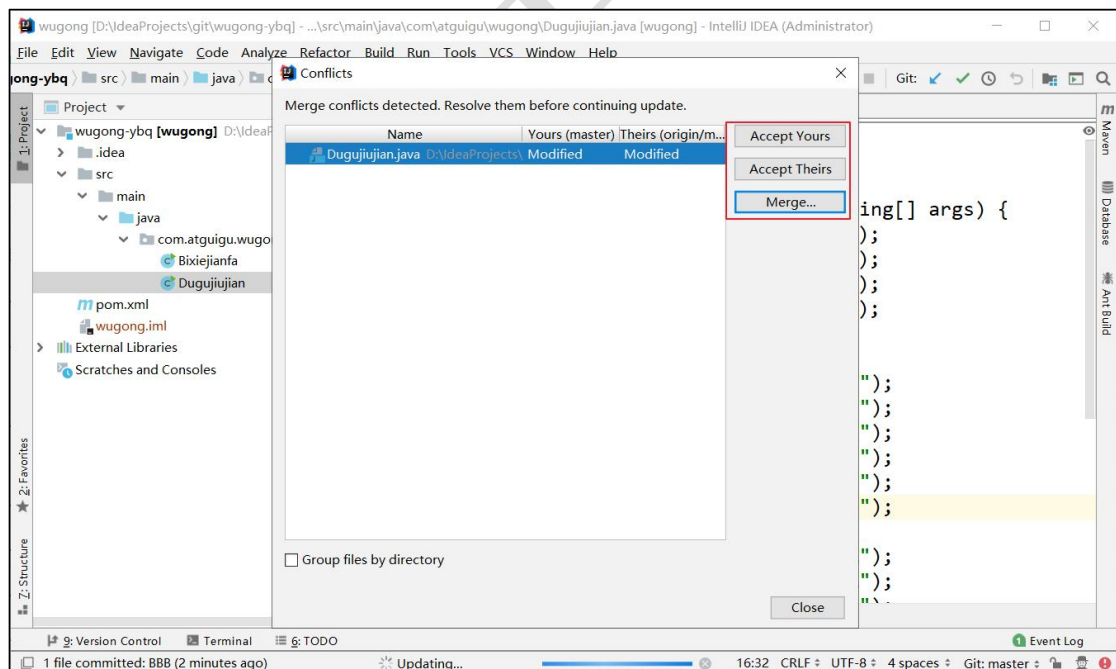
2) 岳不群修改同一个文件进行 commit push



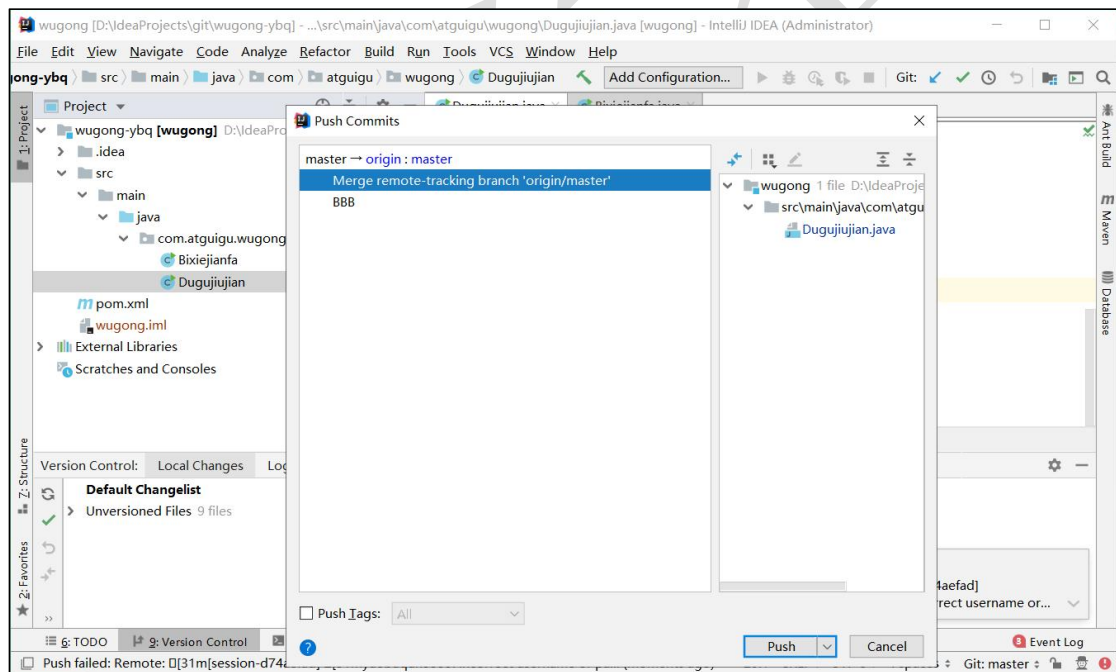
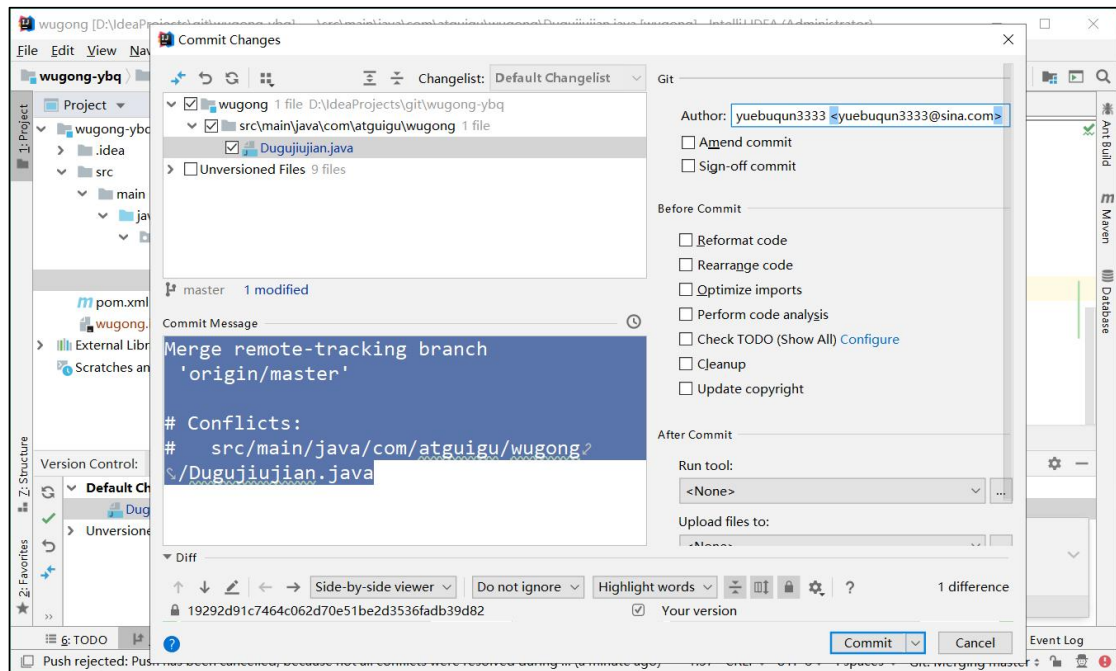
3) 发生冲突

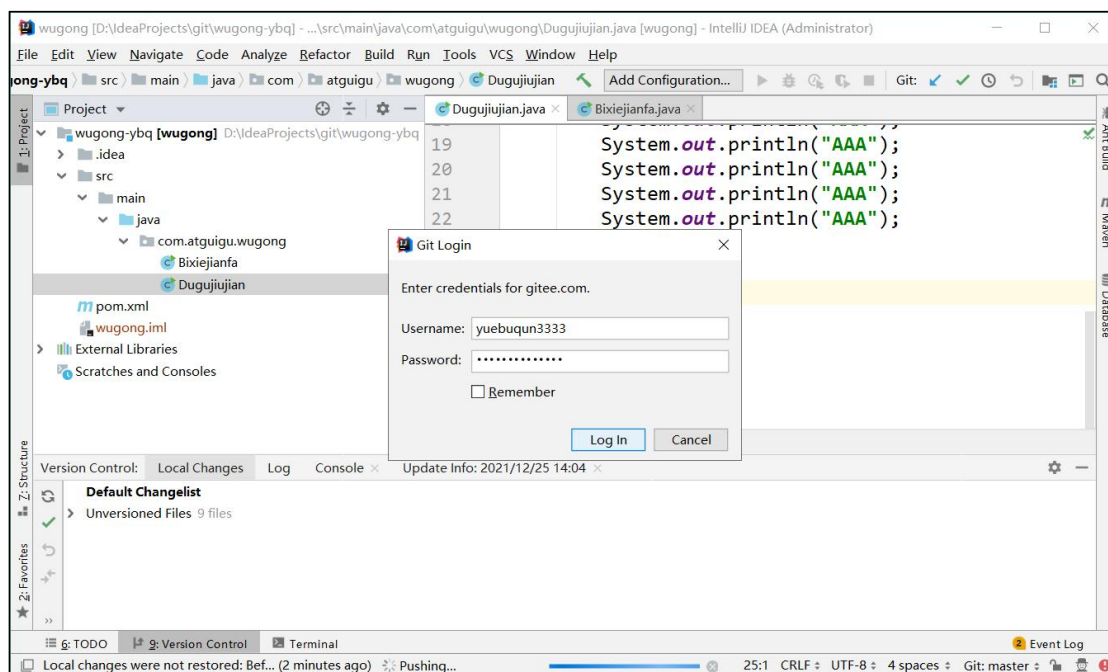


4) 解决冲突









第 6 章 Git workflow

6.1 什么是 Git workflow

简单来说就是，一个项目的成员们在工作中 统一使用 Git 的工作方式

6.2 集中式 workflow

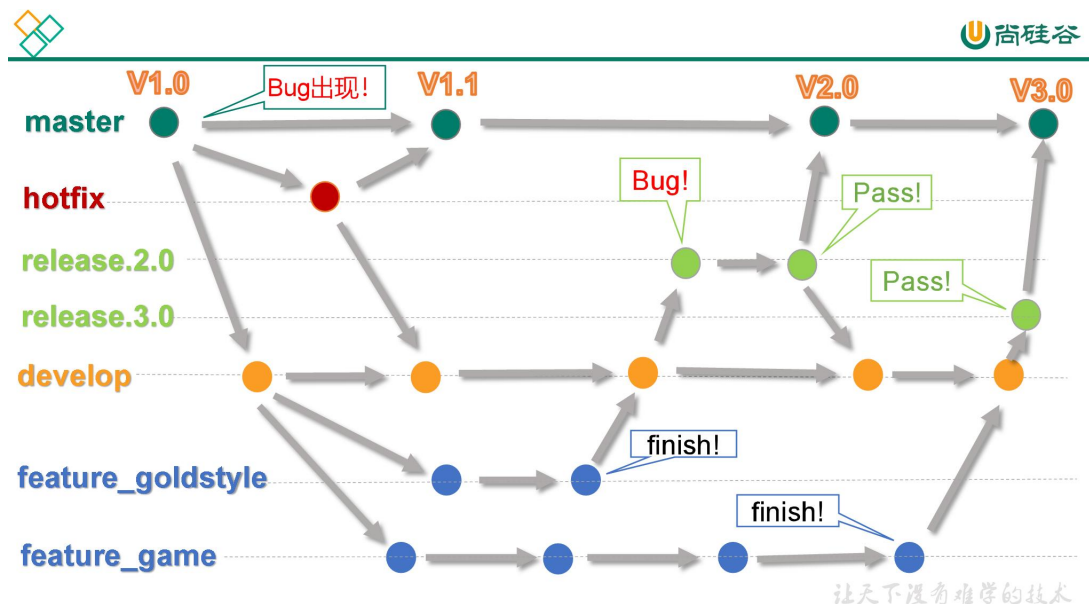
像 SVN 一样，集中式 workflow 以中央仓库作为项目所有修改的单点实体。所有修改都提交到 Master 这个分支上。

这种方式与 SVN 的主要区别就是开发人员有本地库。Git 很多特性并没有用到。

6.3 GitFlow workflow

Gitflow workflow 通过为功能开发、发布准备和维护设立了独立的分支，让发布迭代过程更流畅。严格的分支模型也为大型项目提供了一些非常必要的结构。

6.4 一个比较复杂的项目



6.5 分支种类

1) 主干分支 **master**

主要负责管理正在运行的生产环境代码。永远保持与正在运行的生产环境完全一致

2) 开发分支 **develop**

主要负责管理正在开发过程中的代码。一般情况下应该是最新的代码。

3) bug 修理分支 **hotfix**

主要负责管理生产环境下出现的紧急修复的代码。从主干分支分出，修理完毕并测试上线后，并回主干分支。并回后，视情况可以删除该分支。

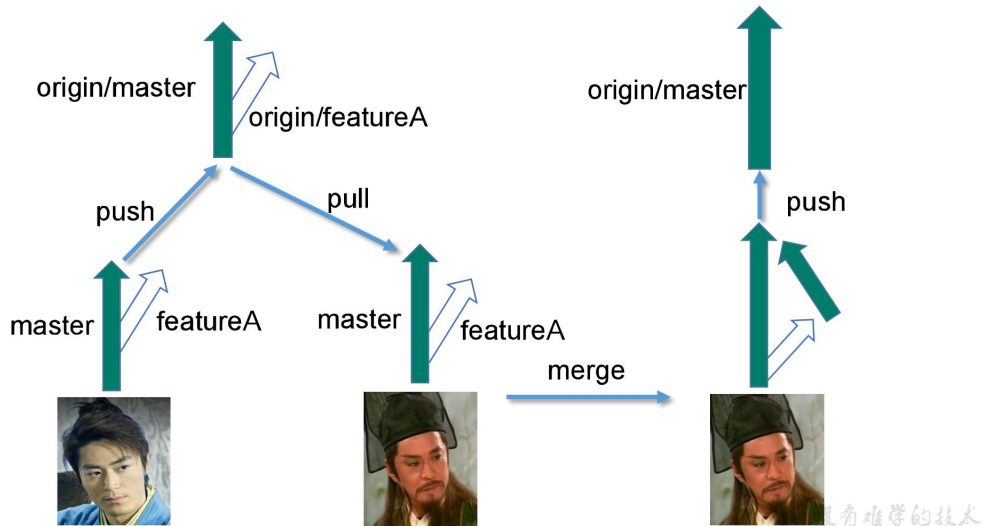
4) 发布版本分支 **release**

较大的版本上线前，会从开发分支中分出发布版本分支，进行最后阶段的集成测试。该版本上线后，会合并到主干分支。生产环境运行一段阶段较稳定后可以视情况删除。

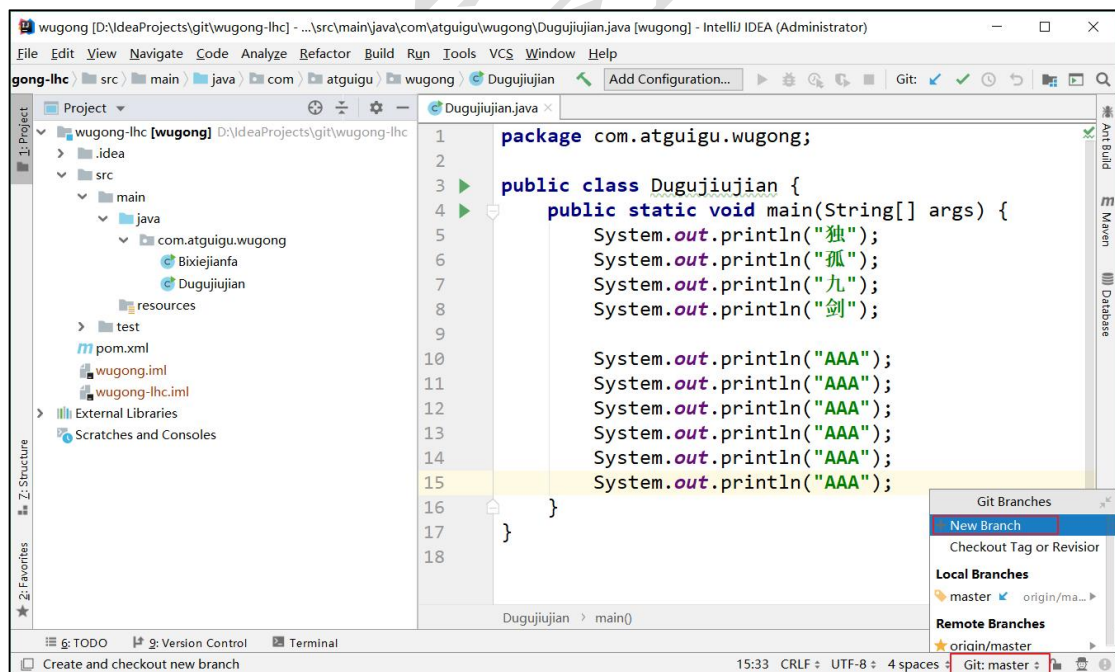
5) 功能分支 **feature**

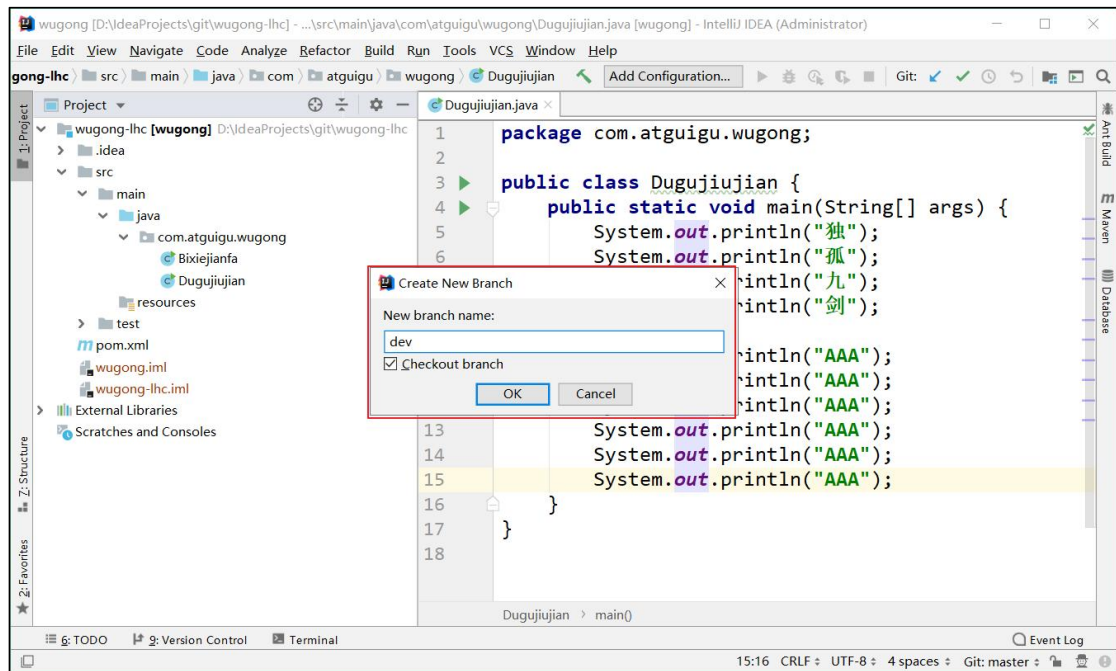
为了不影响较短周期的开发工作，一般把中长期开发模块，会从开发分支中独立出来。开发完成后会合并到开发分支。

6.6 分支实战

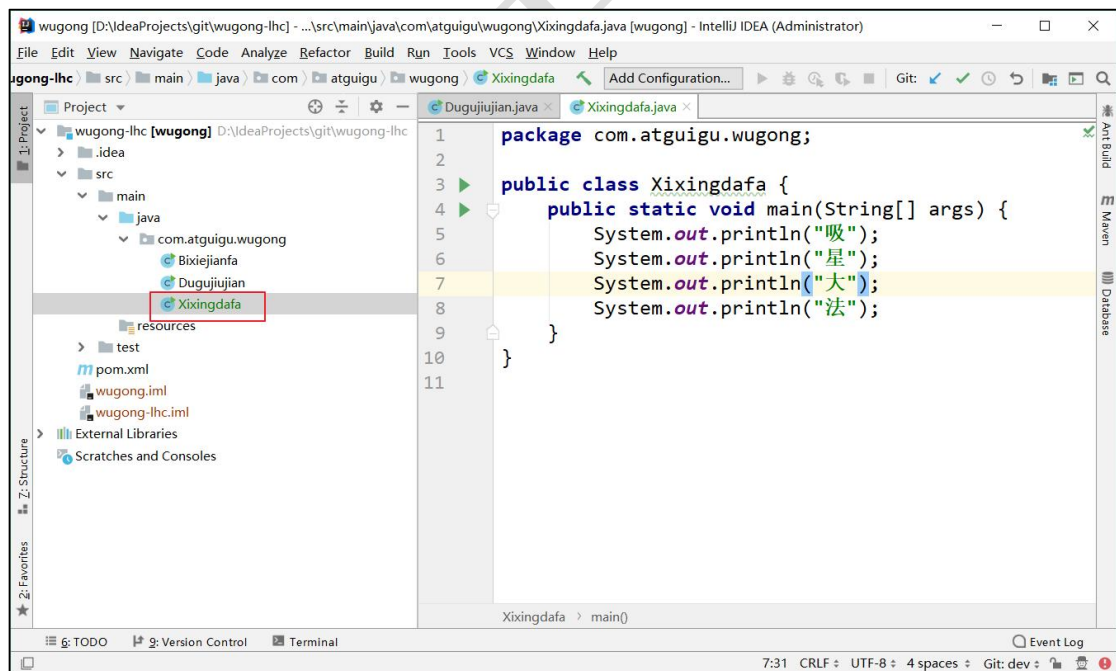


1) 令狐冲 创建一个新的分支 dev

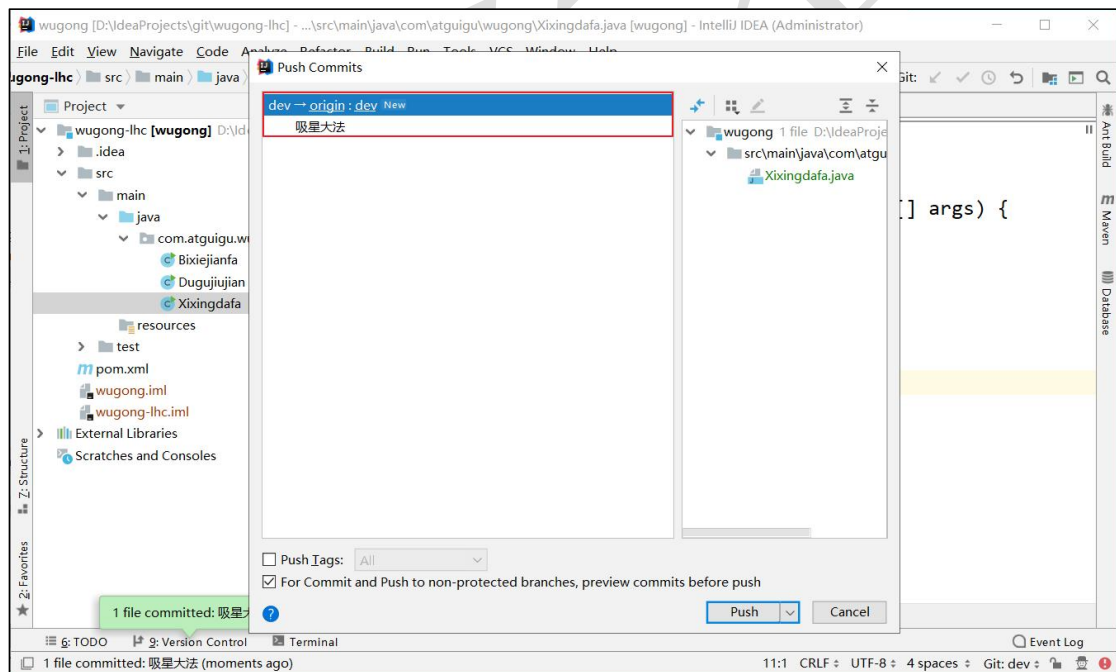
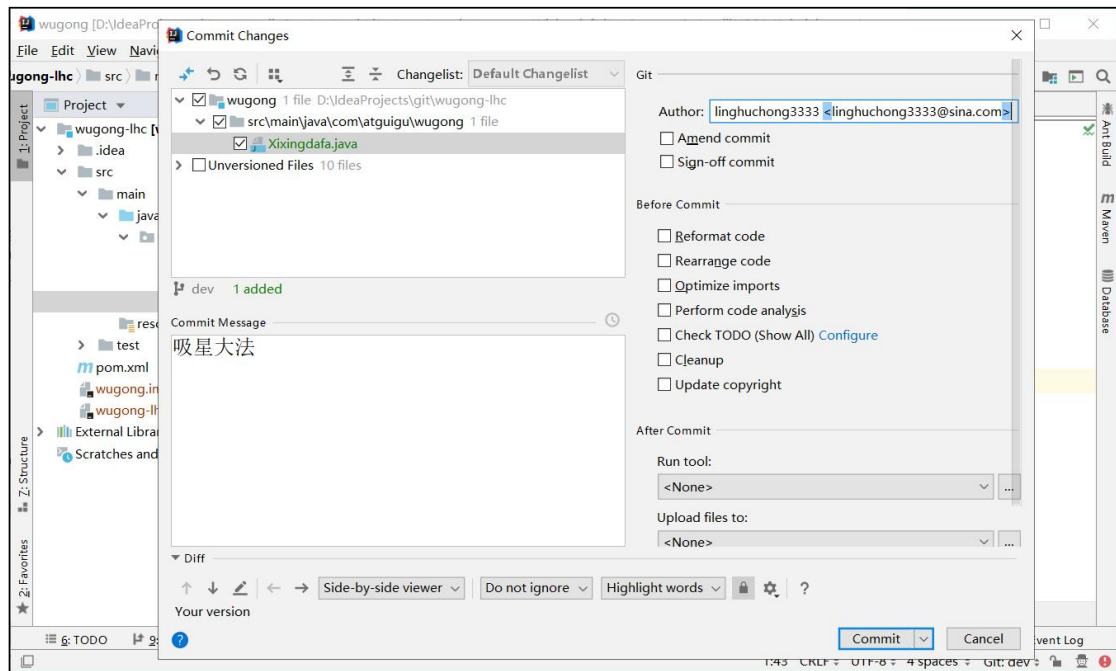


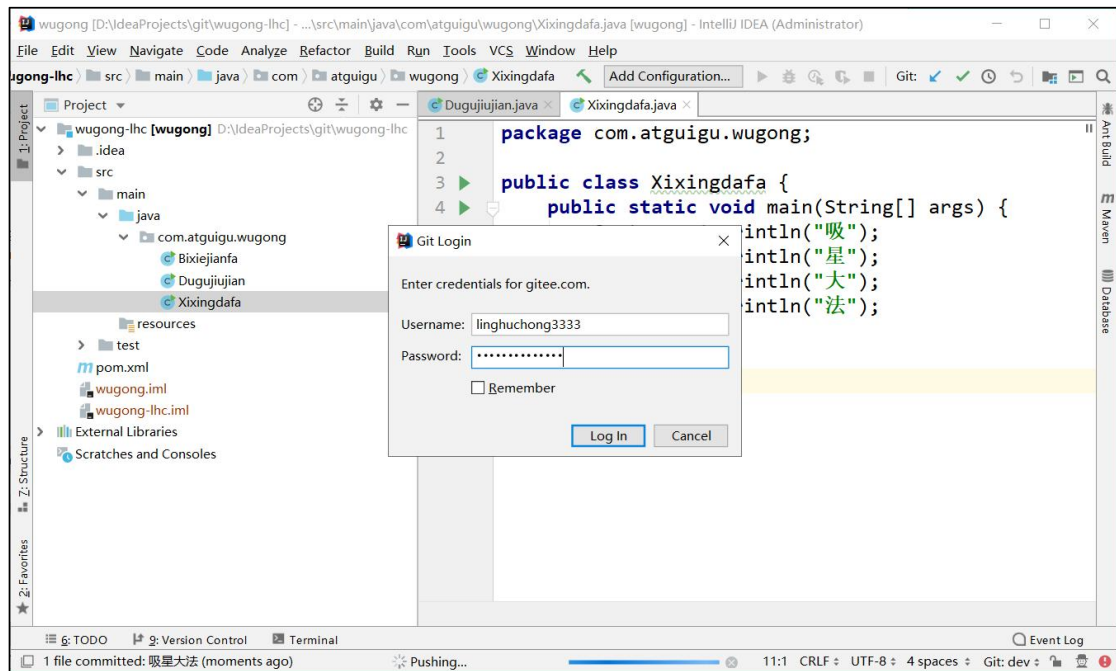


2) 在 dev 分支中进行开发

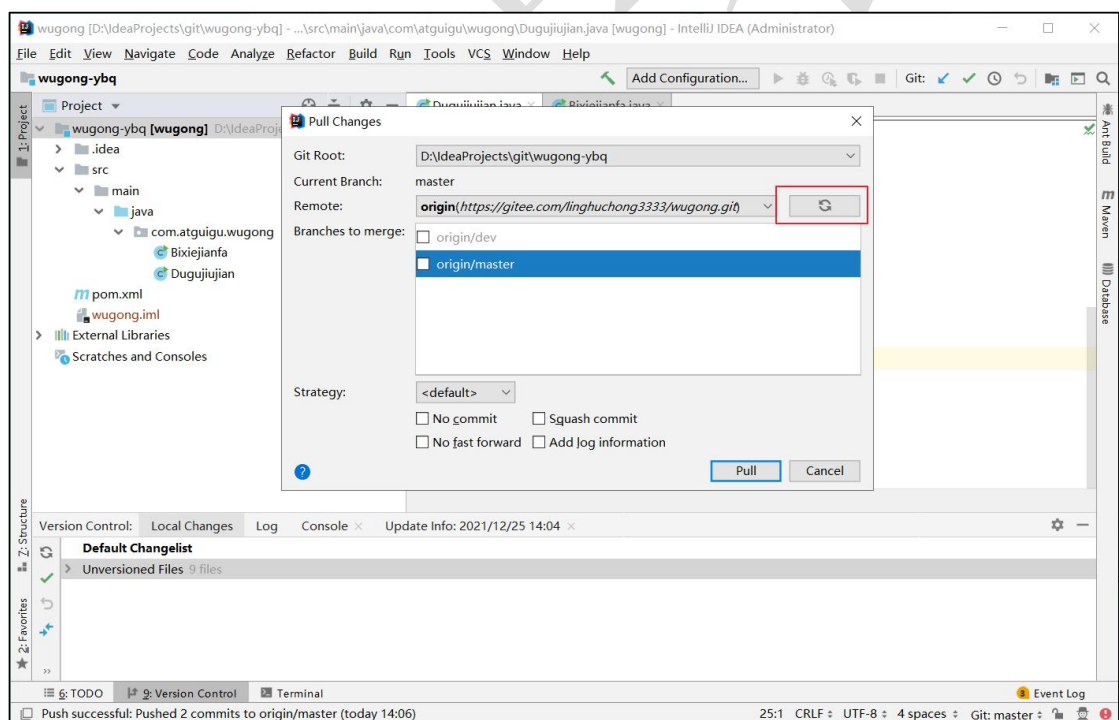
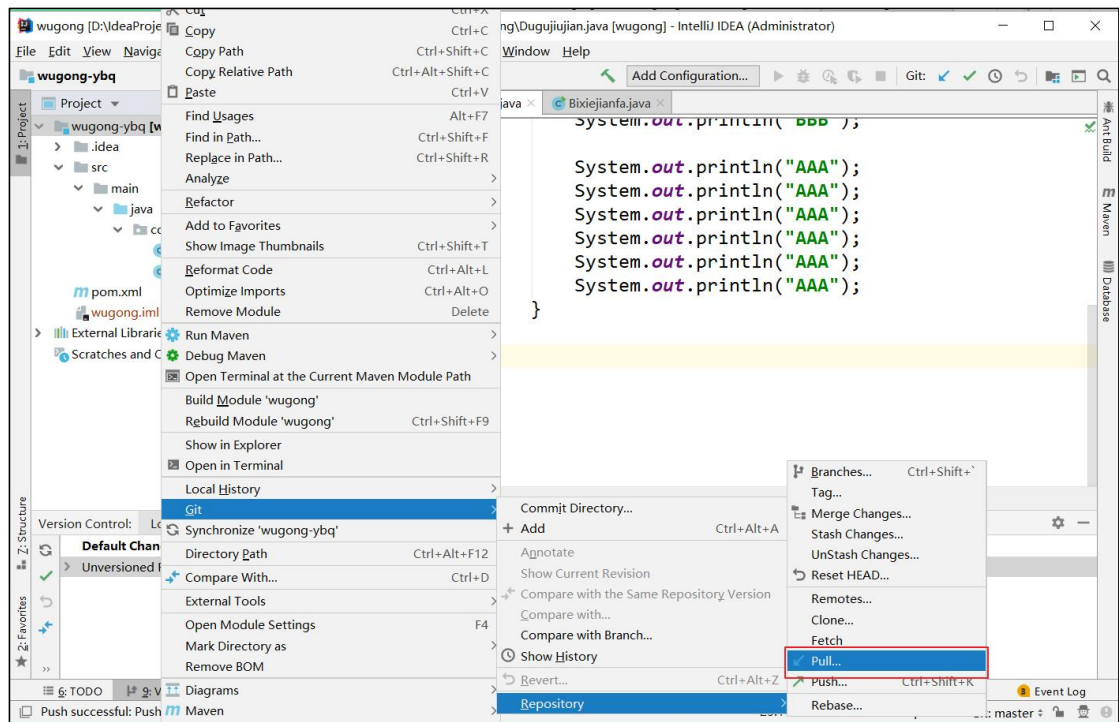


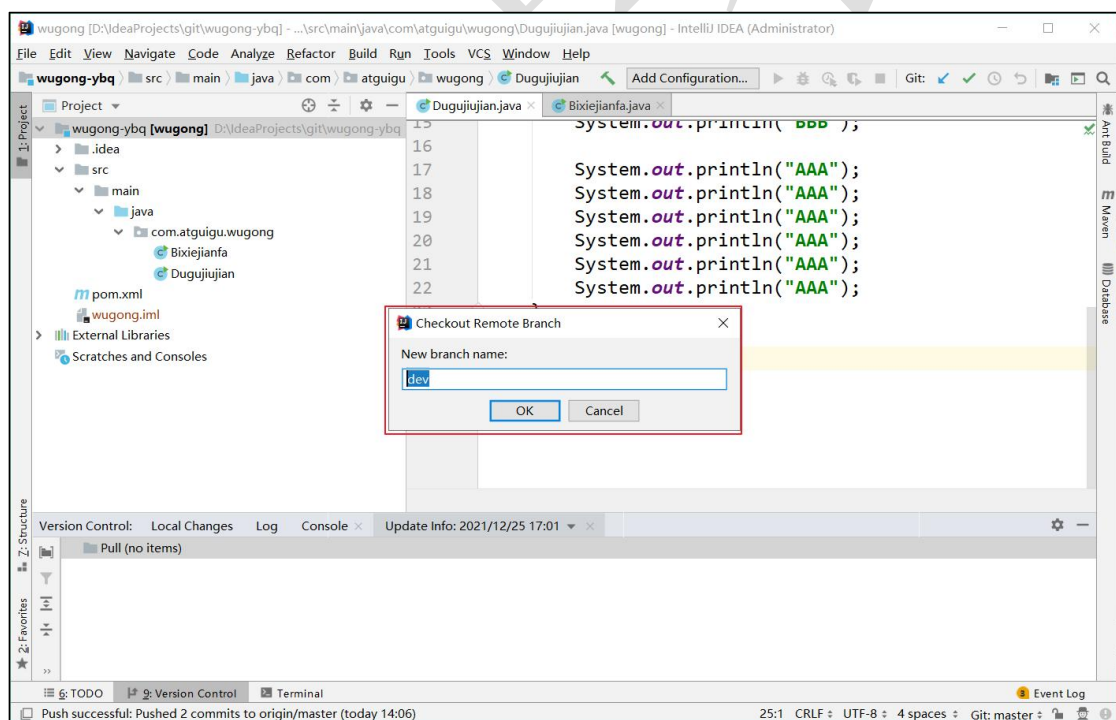
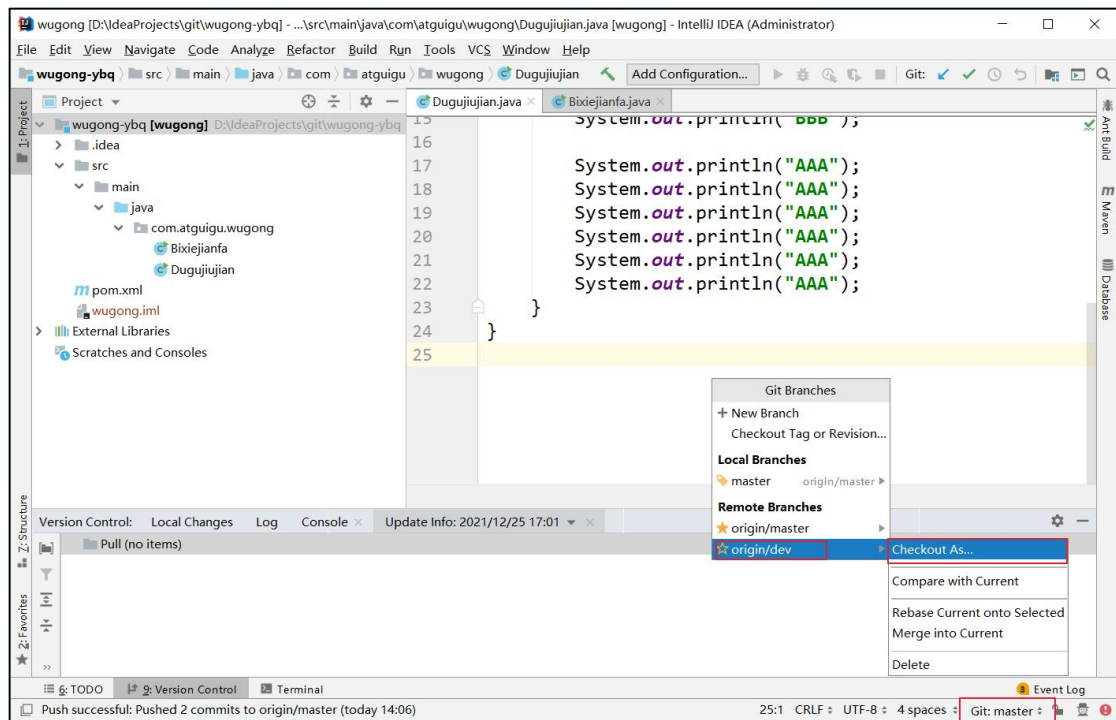
3) 开发完成后，进行 add commit push 的操作





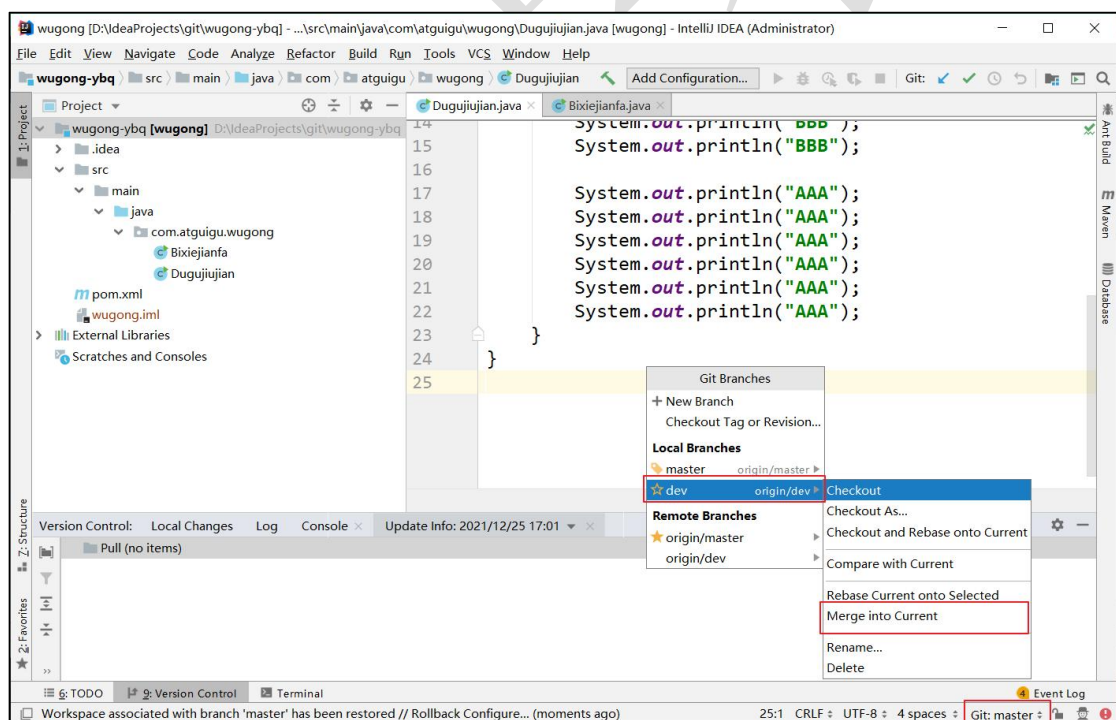
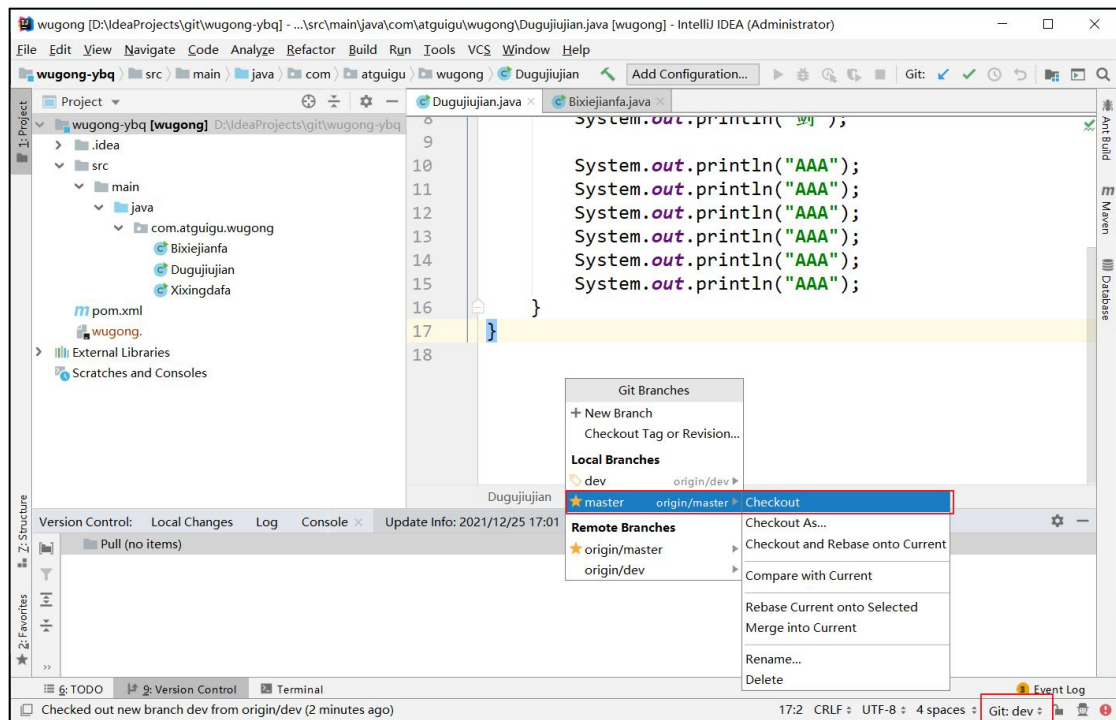
4) 岳不群从 Gitee 上将新的分支取到本地



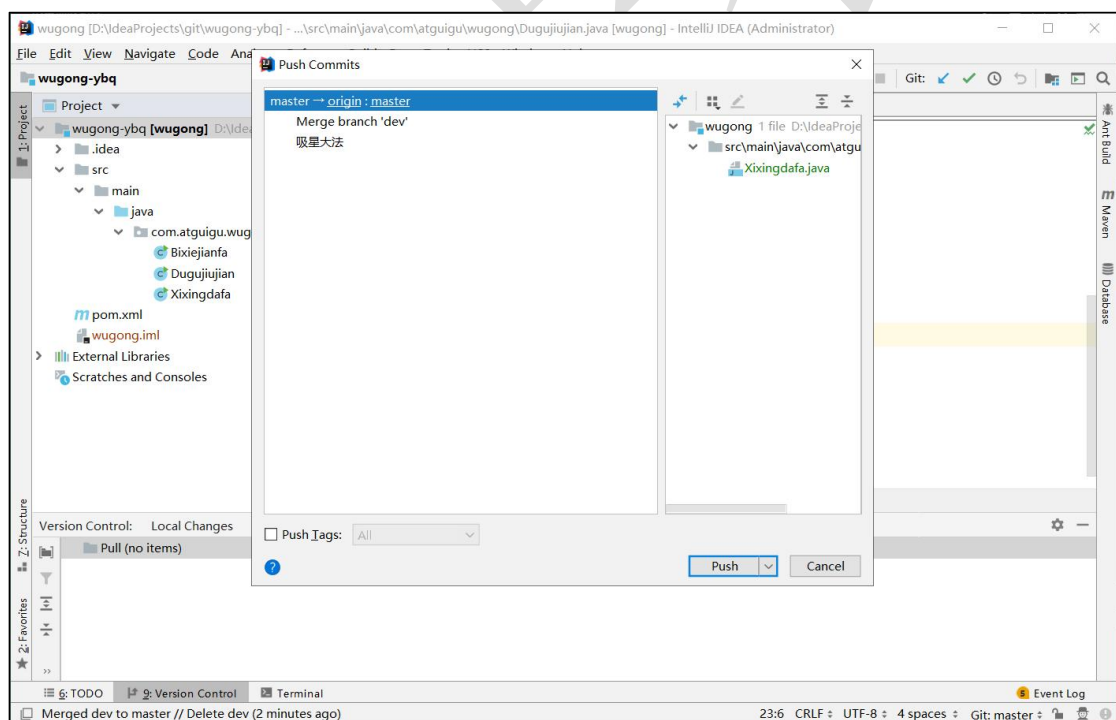
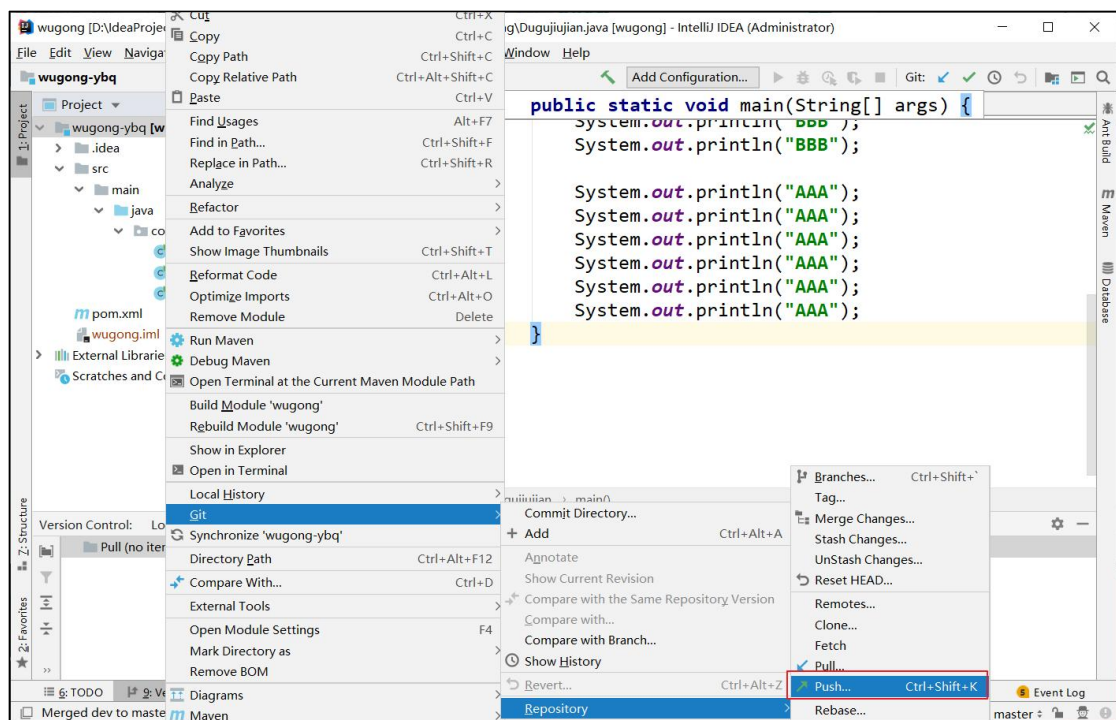


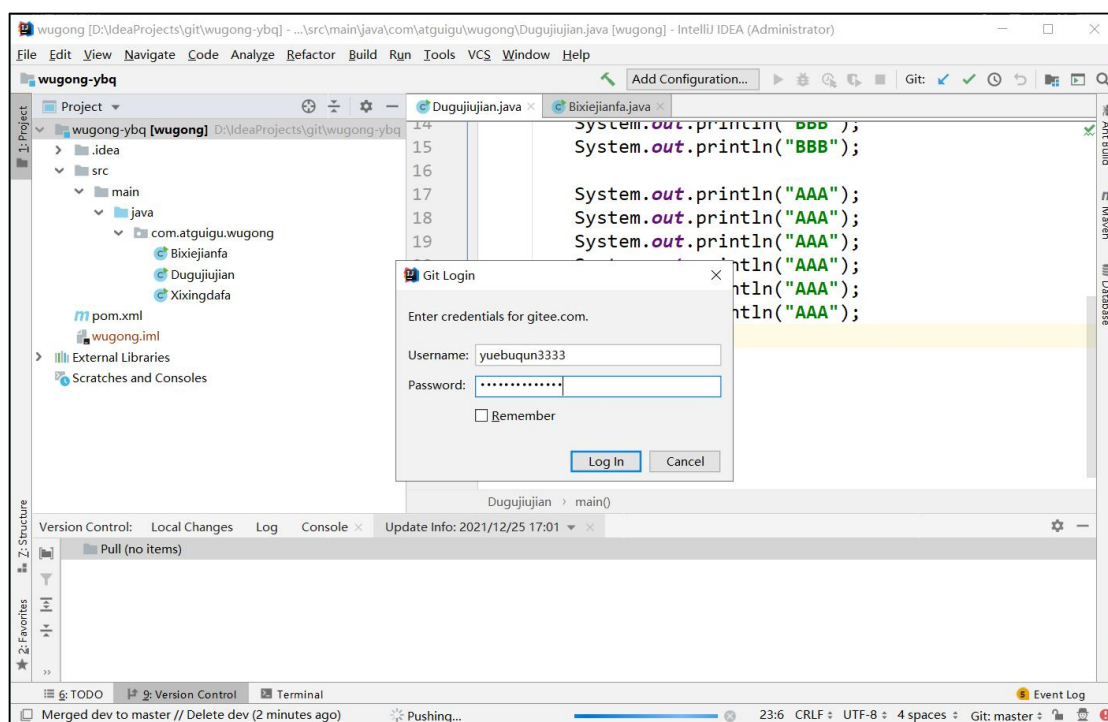
5) 审查分支的代码， 没有问题的情况下，本地 merge

切换到 master 分支上进行合并



6) push





7) 远端查看



第 7 章 SSH 配置

7.1 说明

在进行远端操作时，使用用户名和密码的方式进行验证是比较麻烦的，Gitee 目前还支持用户名和密码的方式，而 GitHub 目前已经不支持使用用户名和密码的方式进行验证。

一般我们会使用 ssh 的方式来代替用户名和密码。

remote: Support for password authentication was removed on August 13, 2021. Please use a personal access token instead.

7.2 配置

1) 生成 ssh 秘钥对文件

在 windows 的 cmd 窗口中生成

```
C:\Users\Administrator> ssh-keygen -t rsa
```

或者是 git 的 bash 窗口都可以生成。

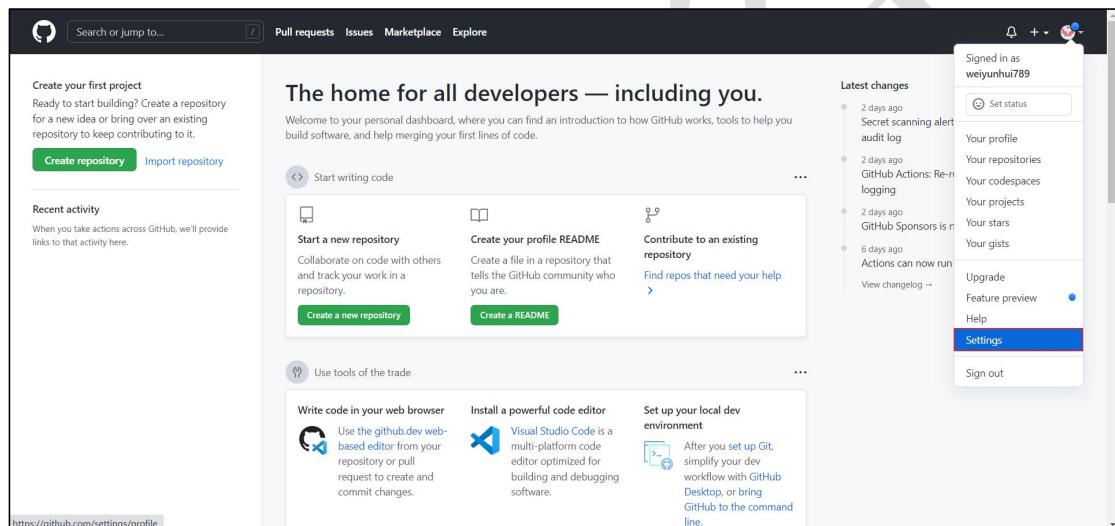
```
$ ssh-keygen -t rsa
```

2) 复制公钥内容

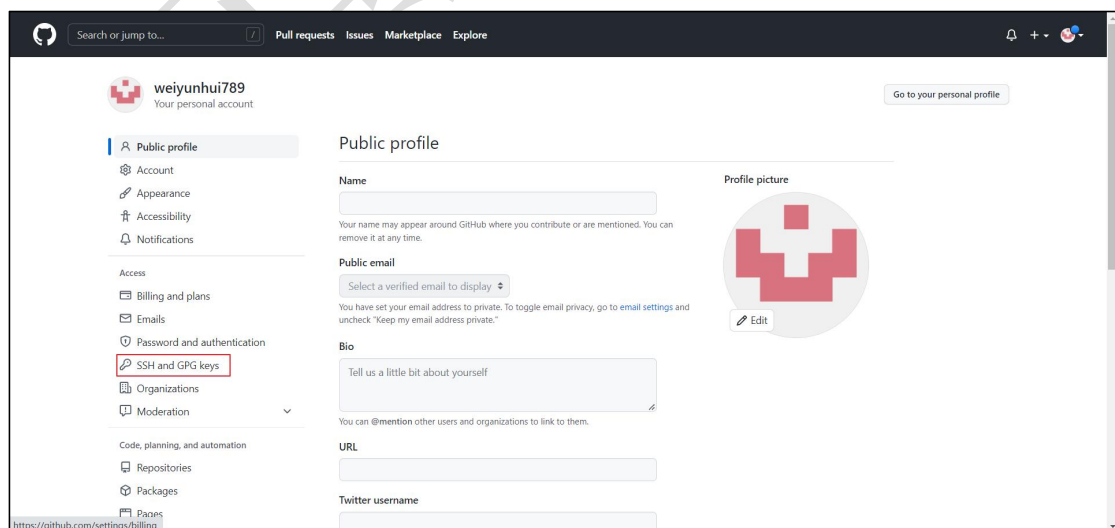
在 C:\Users\当前登录用户\.ssh 目录下找到生成的秘钥对文件,复制 id_rsa.pub 文件的内容

3) 在 github 中配置

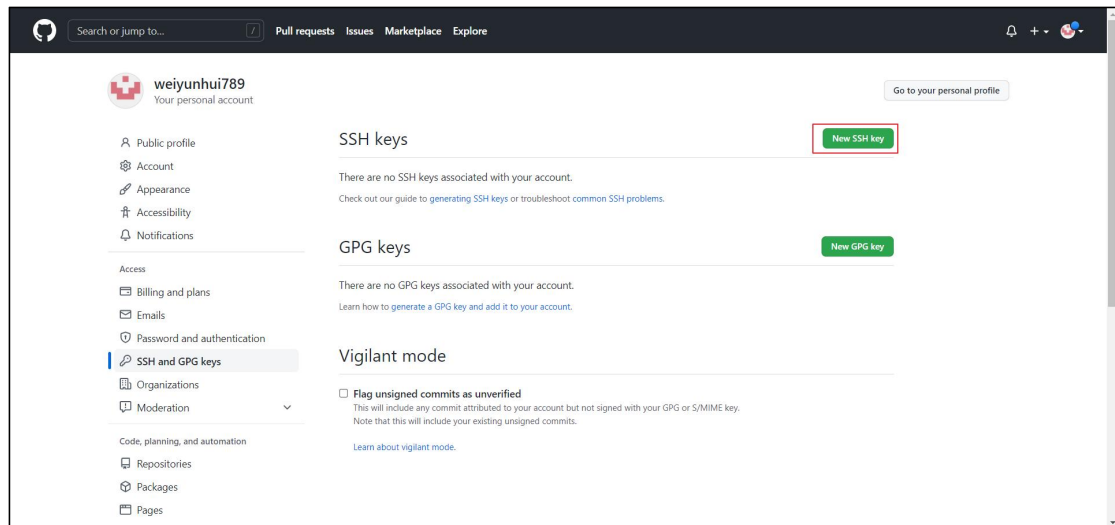
点击用户下的 settings



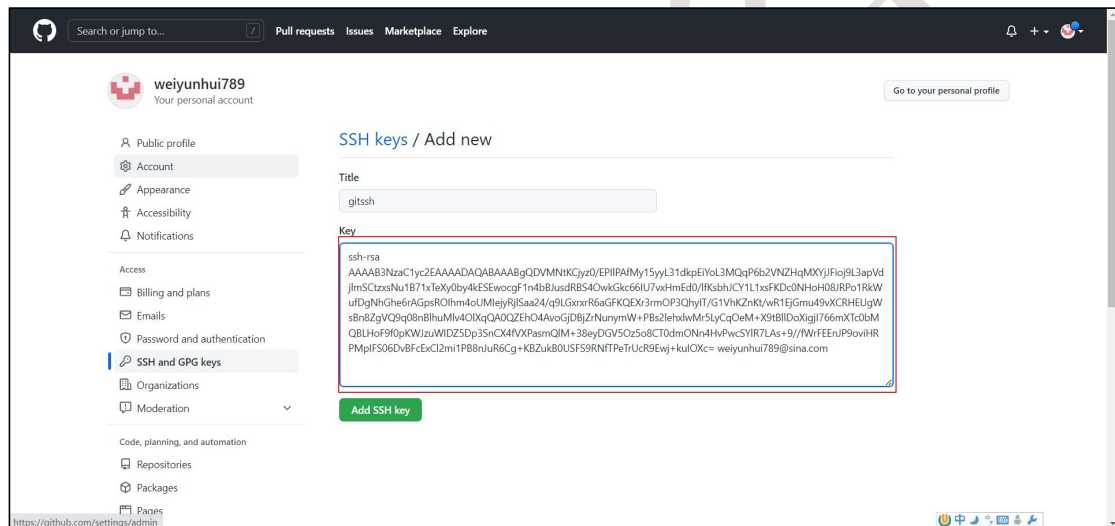
点击 SSH and GPG keys



点击 new SSH key

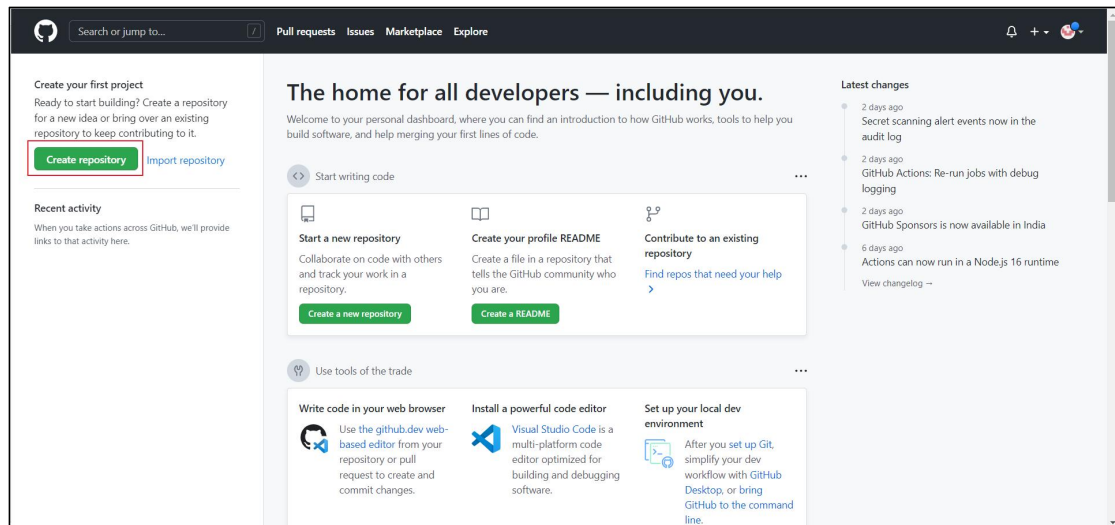


将复制的公钥内容填入后，点击 **Add SSH key** 完成配置

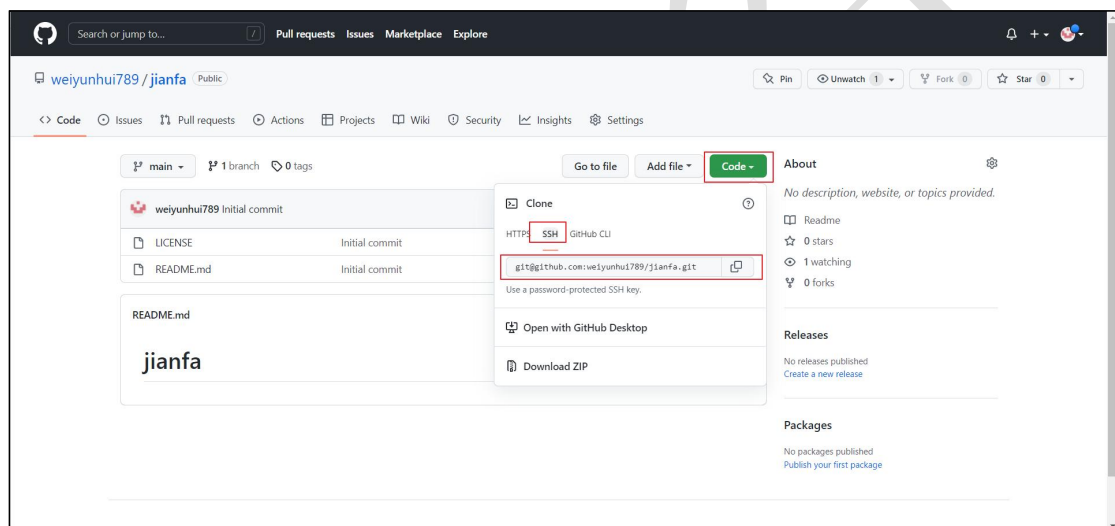


7.3 基本操作

1) 在 github 中创建仓库



2) 复制仓库的 ssh 地址



3) 在本地仓库配置远端代号并进行 push

```
$ git remote add origin git@github.com:weiyunhui789/jianfa.git
$ git push origin master
```